



Genomics of Gene Regulation: Concepts and Techniques

Ana Azambuja

Senior Research Associate

Pathology Department - Boston Children's Hospital



Genomics of Gene Regulation: Concepts and **Techniques**

Analysis of gene expression and gene targeting

Epigenomic regulation and chromatin architecture

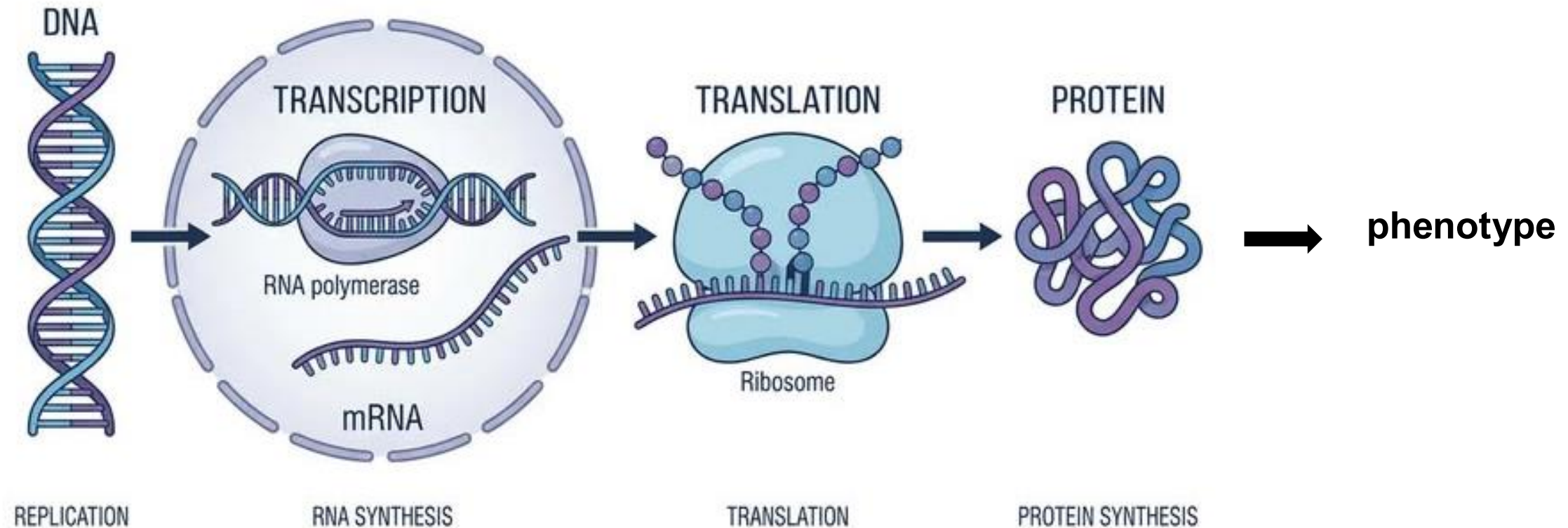
Enhancer analysis and targeting

Ana Azambuja

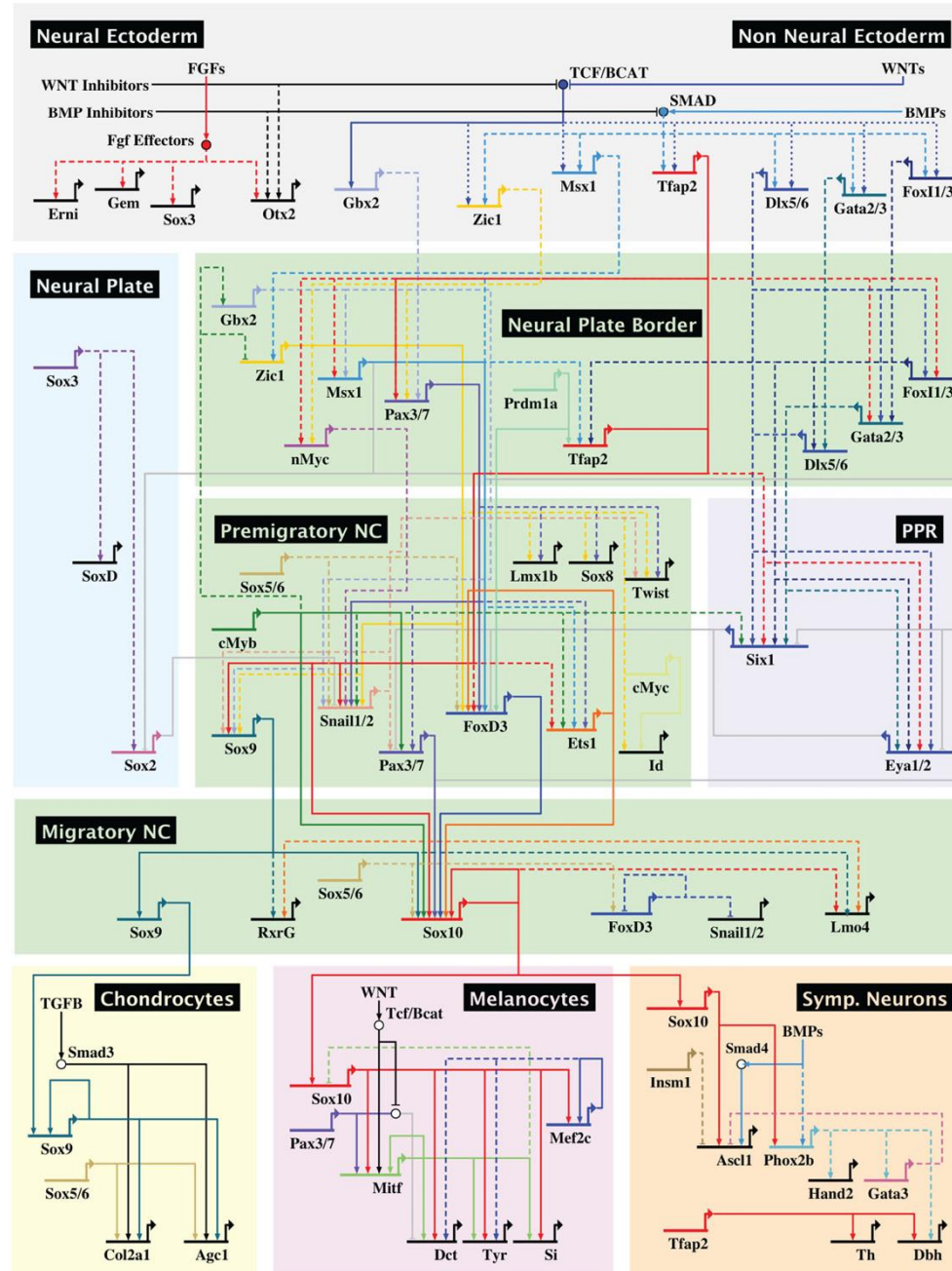
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The information flux in biological systems



Phenotypes are the product of the interaction of different molecules



transcription factors

structural proteins

enhancers

repressors

splicing machinery

chromatin modifiers

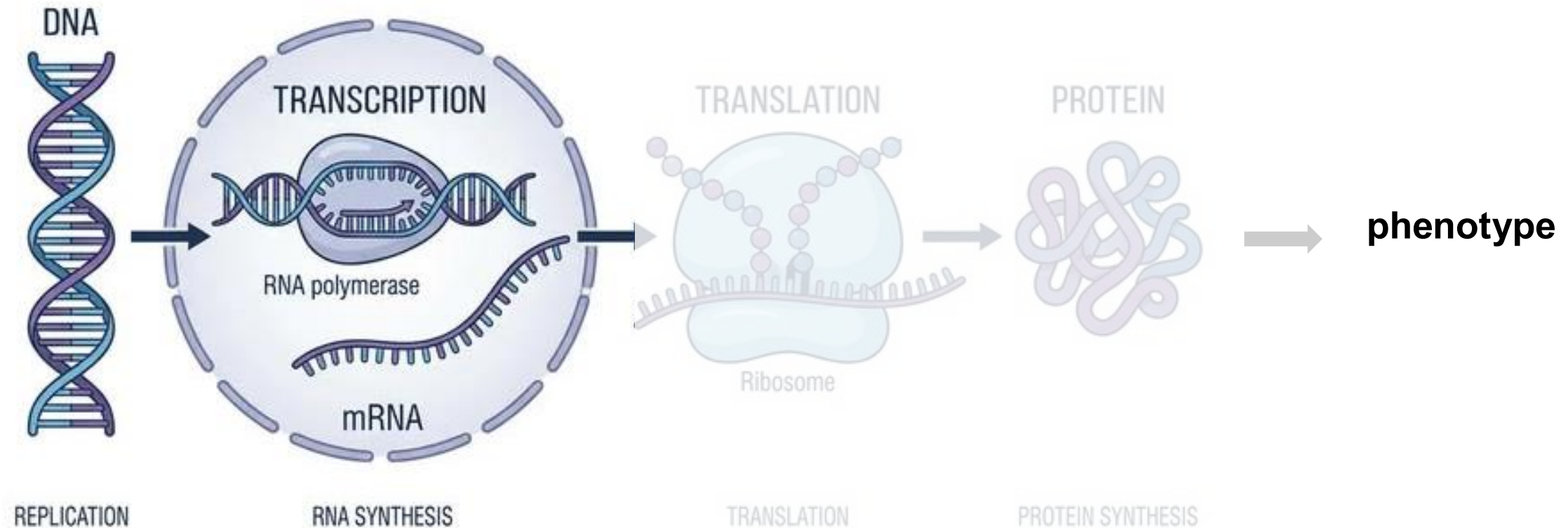
lncRNA

metabolites

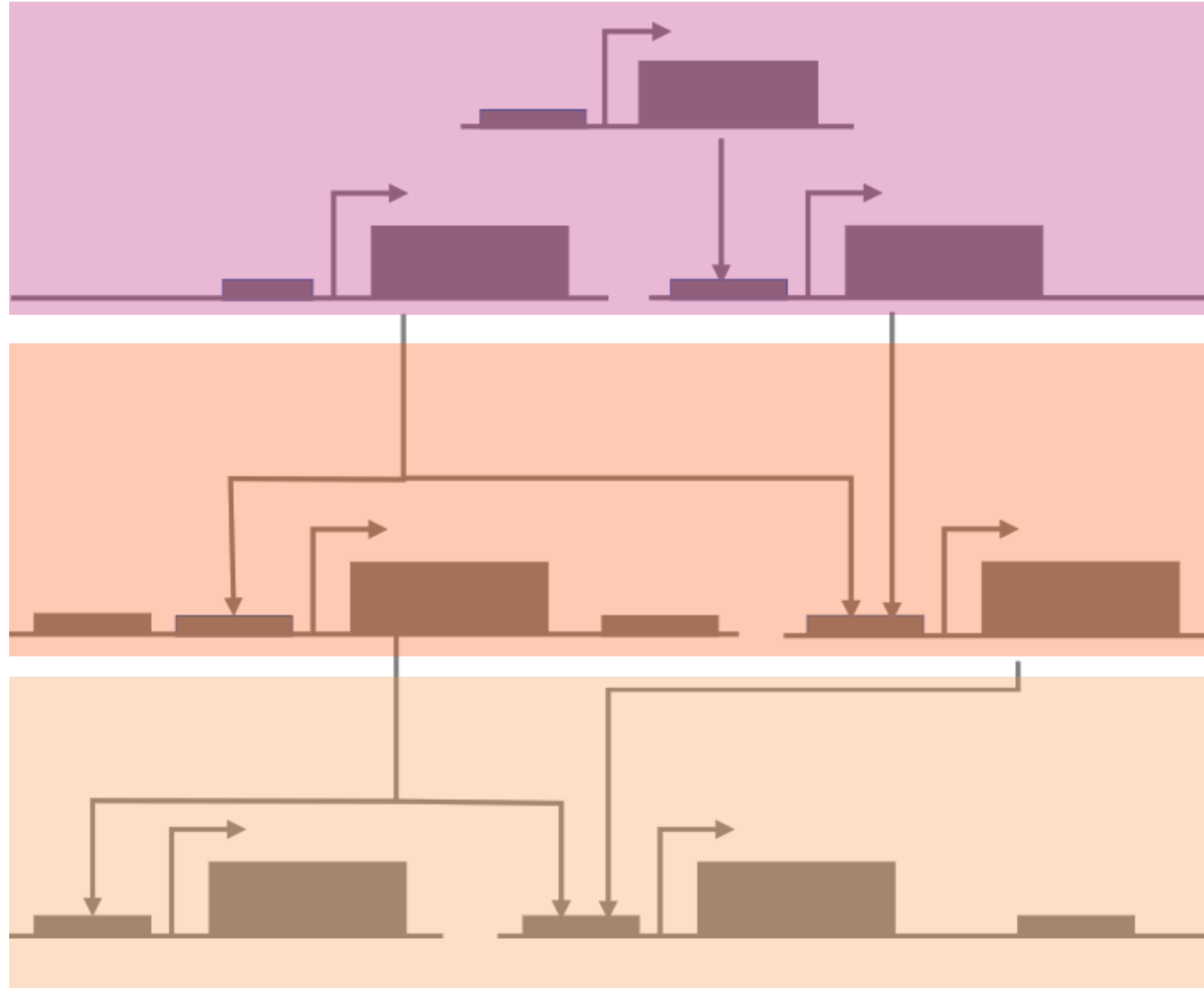
co-factors

signaling molecules

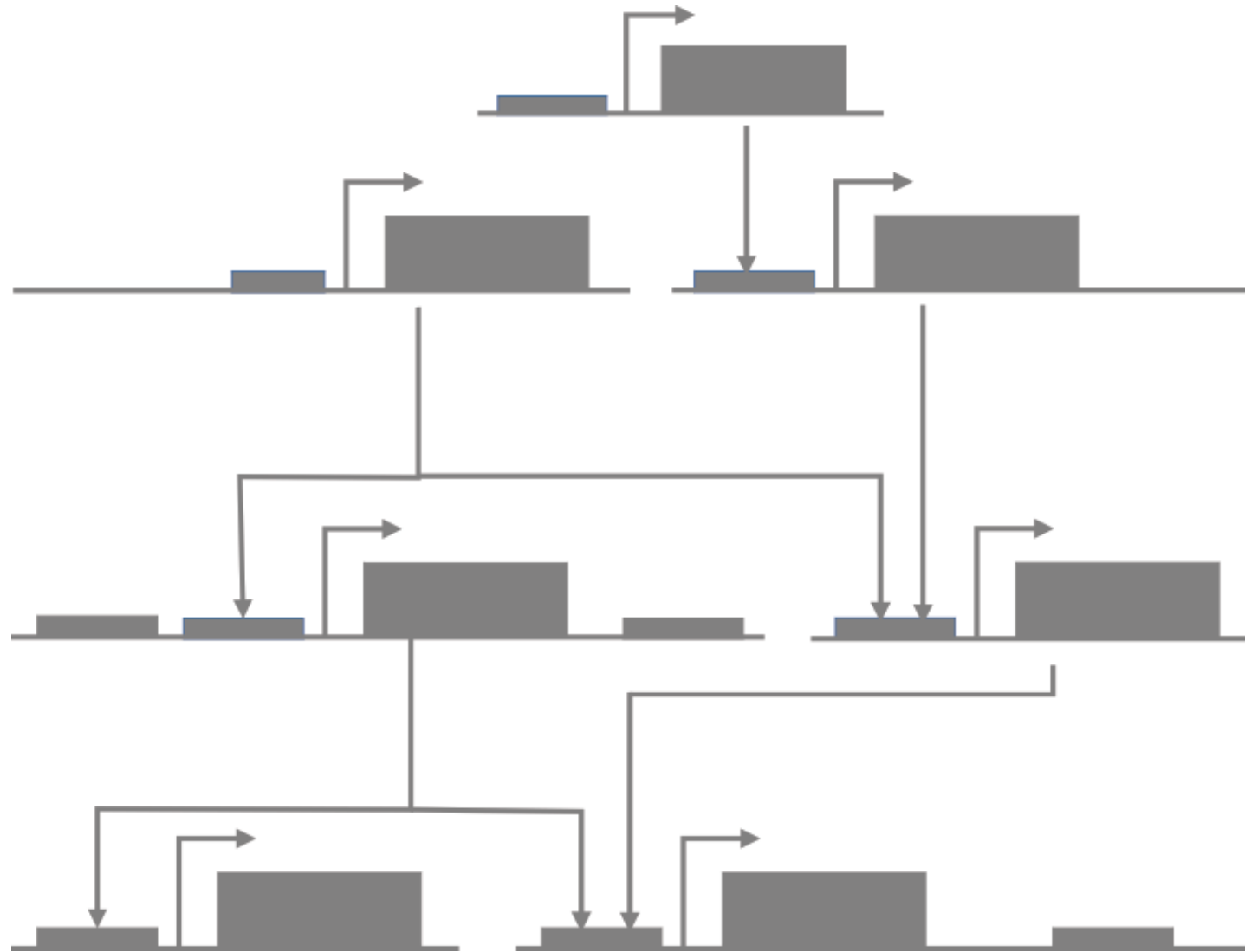
The information flux in biological systems



Cellular transitions are controlled by GENE REGULATORY NETWORKS



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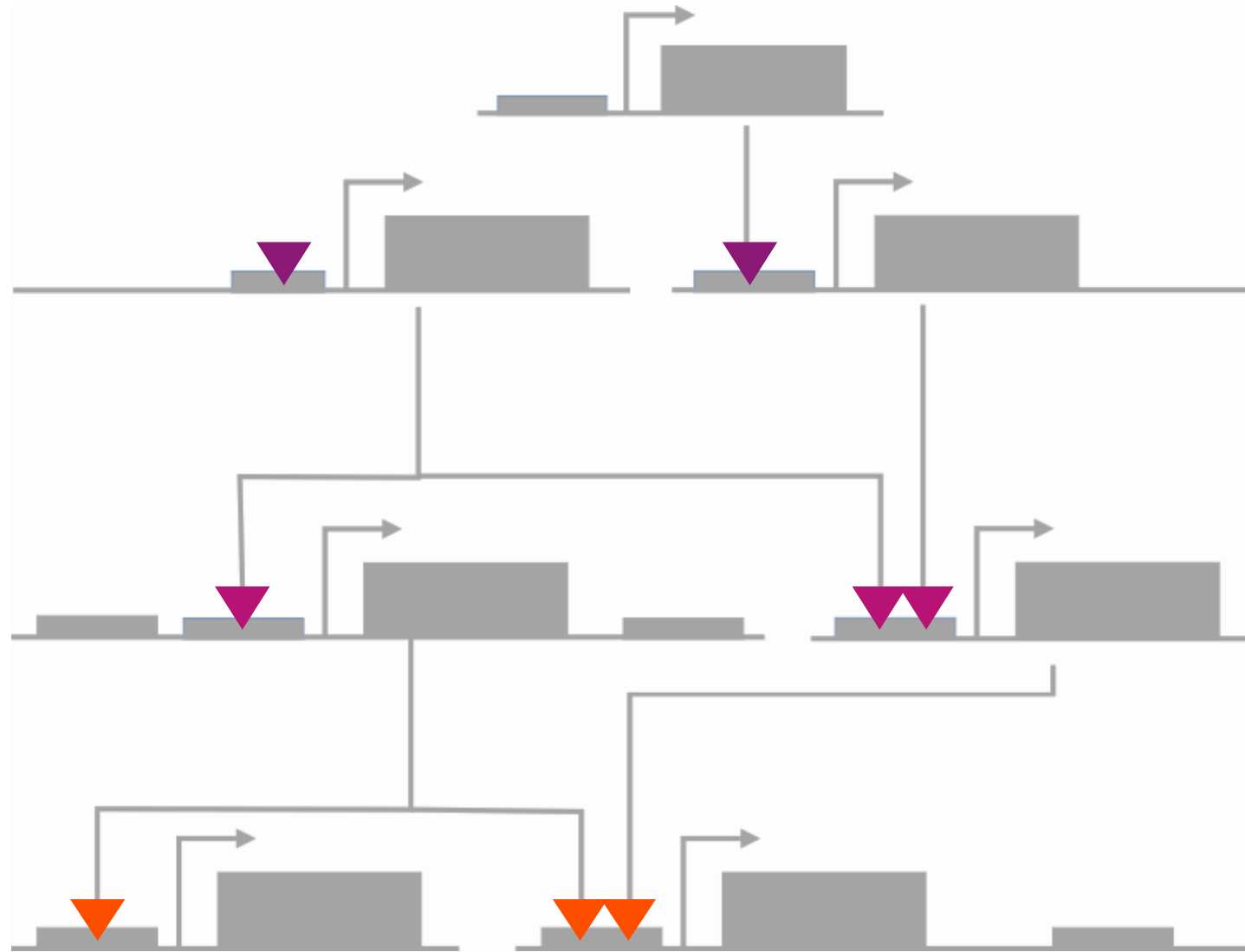


Marker genes

Cellular transitions are controlled by GENE REGULATORY NETWORKS

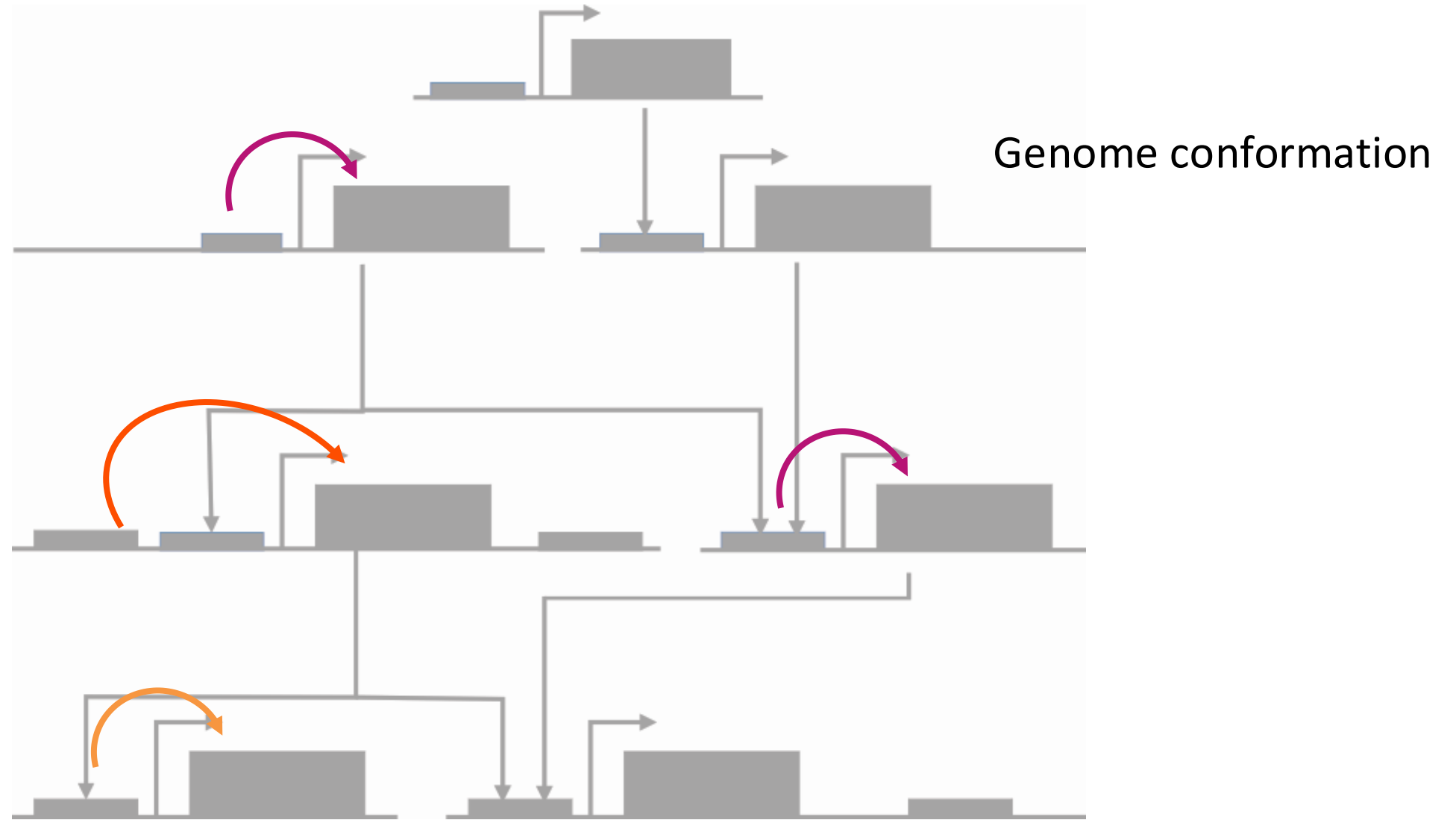


Cellular transitions are controlled by GENE REGULATORY NETWORKS



Protein –
DNA interactions

Cellular transitions are controlled by GENE REGULATORY NETWORKS

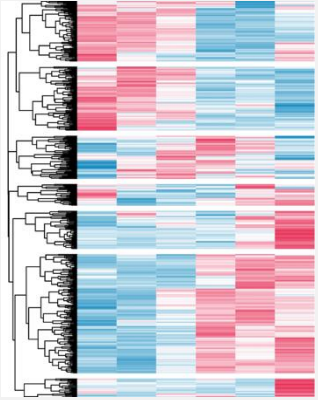


Topics to be covered - Techniques

GENE EXPRESSION

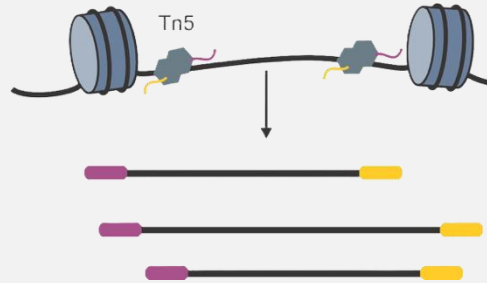
RNAseq

Spatial transcriptomics



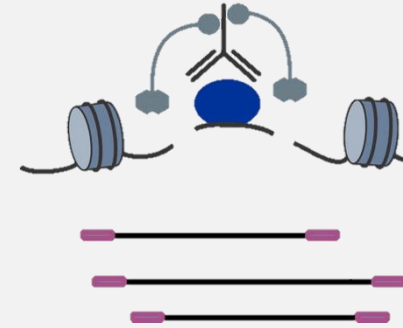
EPIGENOMIC REGULATION

ATACseq



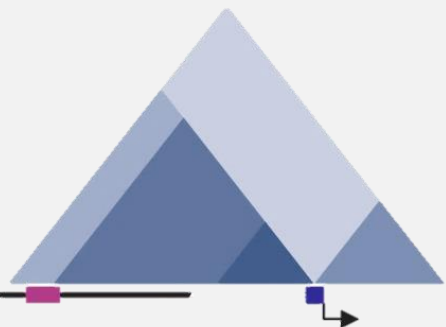
DNA-PROTEIN BINDING

ChIPseq
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CUT&TAG



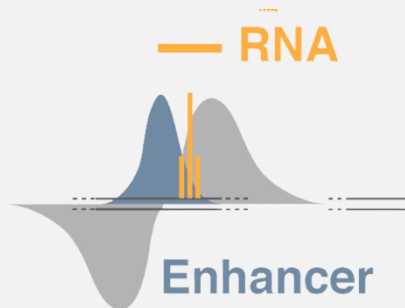
GENOME ORGANIZATION

3D genome organization - HiC



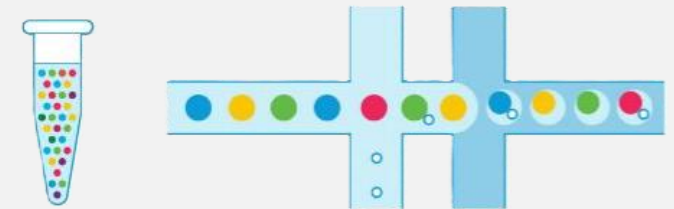
ENHANCER ACTIVITY

eRNA mapping

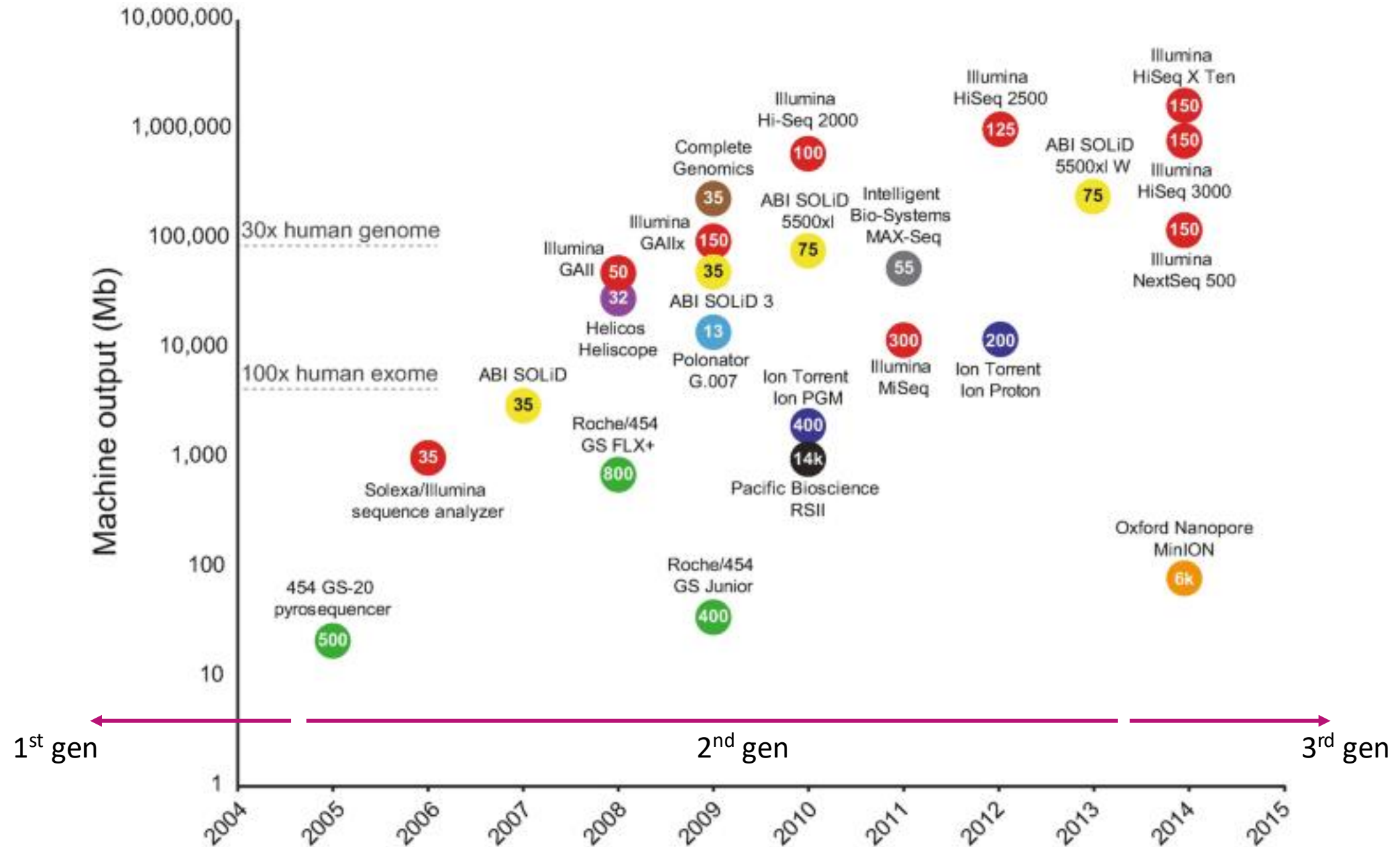


SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome



High Throughput Sequencing evolution



High Throughput Sequencing - Platforms

2nd Gen (short reads) vs. 3rd Gen (long reads)

Illumina

Oxford Nanopore

sequencing by synthesis

highly accurate base-by-base sequencing

short reads / paired-end sequencing

(up to 600bp) large-scale sequencing

cluster formation (solid-phase amplification)

accurate base calls defined from signal intensity

applications

RNAseq, HiC, CUT&RUN, ChIP...

real-time sequencing

adaptive sampling available

no read length restrictions

from short to ultra-long (>4 Mb)

amplification independent

more accurate identification of base modifications,
GC rich and highly repetitive regions

applications

genome assembly, structural variants...

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Long Read / Single Molecule Sequencing – Oxford Nanopore

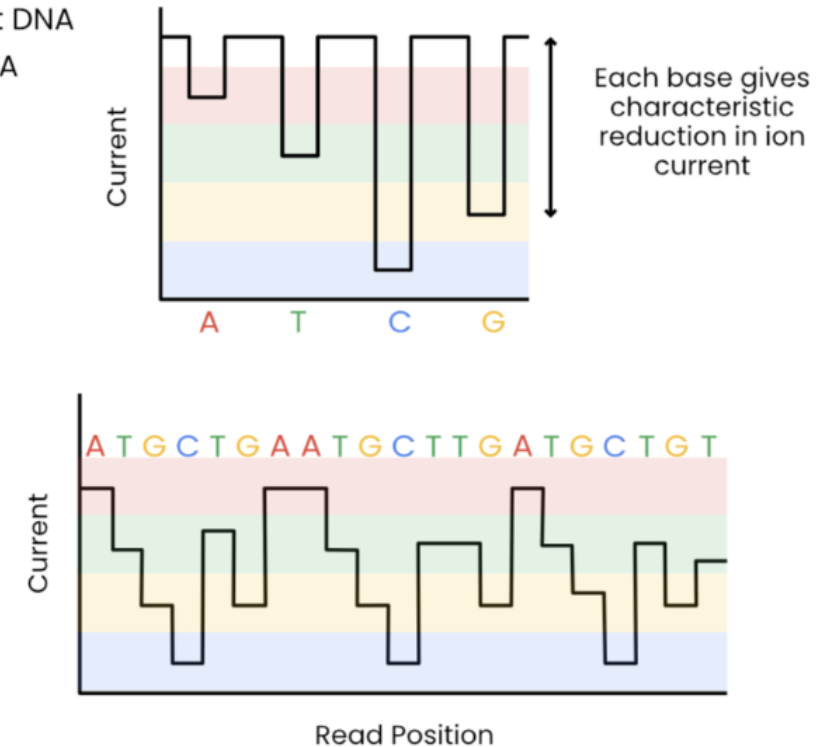
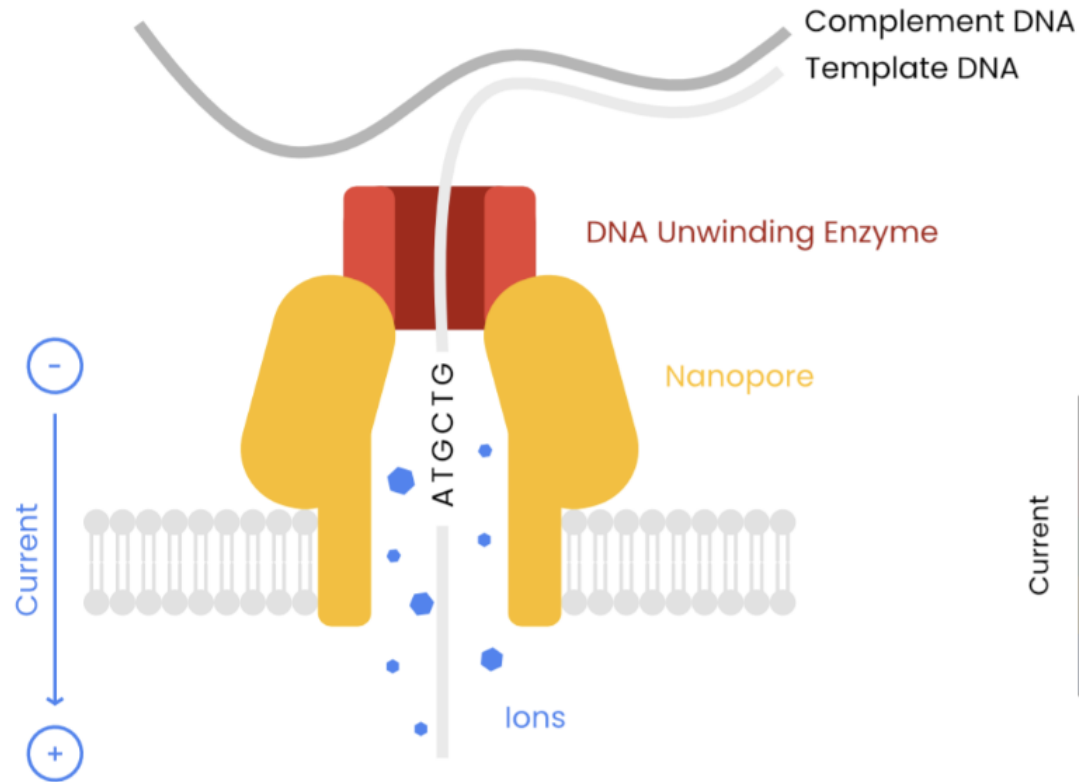
RESEARCH ARTICLE | BIOPHYSICS AND COMPUTATIONAL BIOLOGY | ✓



Characterization of individual polynucleotide molecules using a membrane channel

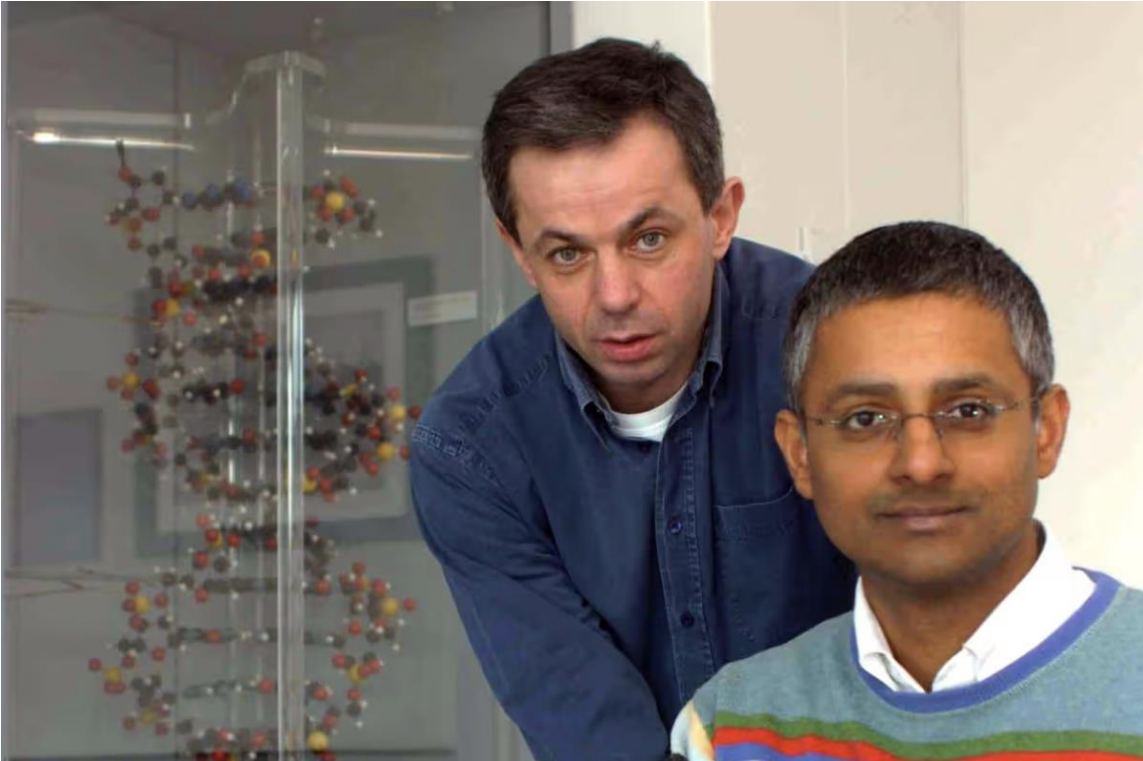
[John J. Kasianowicz](#), [Eric Brandin](#), [Daniel Branton](#), and [David W. Deamer](#) [Authors Info & Affiliations](#)

November 26, 1996 | 93 (24) 13770-13773 | <https://doi.org/10.1073/pnas.93.24.13770>

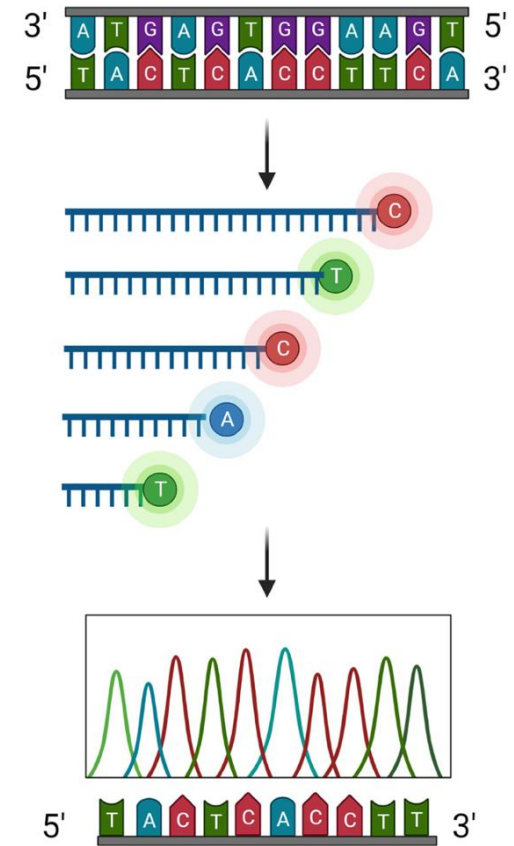


Short Read Sequencing – Illumina Platforms

How to watch a single DNA polymerase molecule at work?



David Klenerman and Shankar Balasubramanian



Sanger Sequencing

ARTICLES

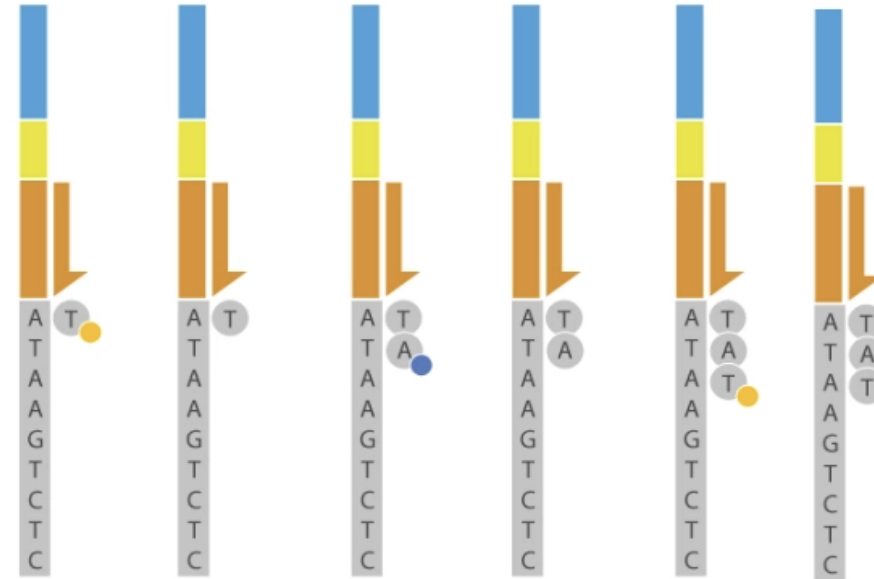
Accurate whole human genome sequencing using reversible terminator chemistry

A list of authors and their affiliations appears at the end of the paper

DNA sequence information underpins genetic research, enabling discoveries of important biological or medical benefit. Sequencing projects have traditionally used long (400–800 base pair) reads, but the existence of reference sequences for the human and many other genomes makes it possible to develop new, fast approaches to re-sequencing, whereby shorter reads are compared to a reference to identify intraspecies genetic variation. Here we report an approach that generates several billion bases of accurate nucleotide sequence per experiment at low cost. Single molecules of DNA are attached to a flat surface, amplified *in situ* and used as templates for synthetic sequencing with fluorescent reversible terminator deoxyribonucleotides. Images of the surface are analysed to generate high-quality sequence. We demonstrate application of this approach to human genome sequencing on flow-sorted X chromosomes and then scale the approach to determine the genome sequence of a male Yoruba from Ibadan, Nigeria. We build an accurate consensus sequence from $>30\times$ average depth of paired 35-base reads. We characterize four million single-nucleotide polymorphisms and four hundred thousand structural variants, many of which were previously unknown. Our approach is effective for accurate, rapid and economical whole-genome re-sequencing and many other biomedical applications.

Short Read Sequencing – Illumina Platforms

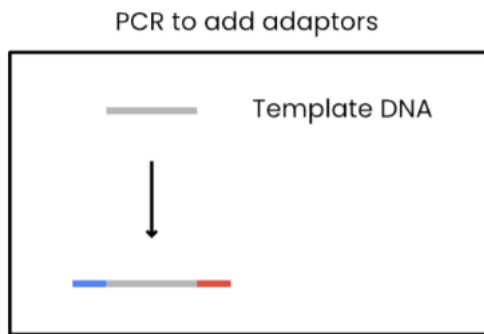
Sequencing by Synthesis (SBS)



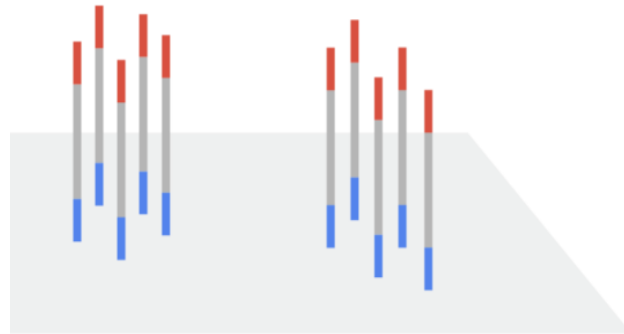
TAT...
CTG...
TGA...

Short Read Sequencing – Illumina Platforms

Sample prep



Cluster Amplification



Sequencing

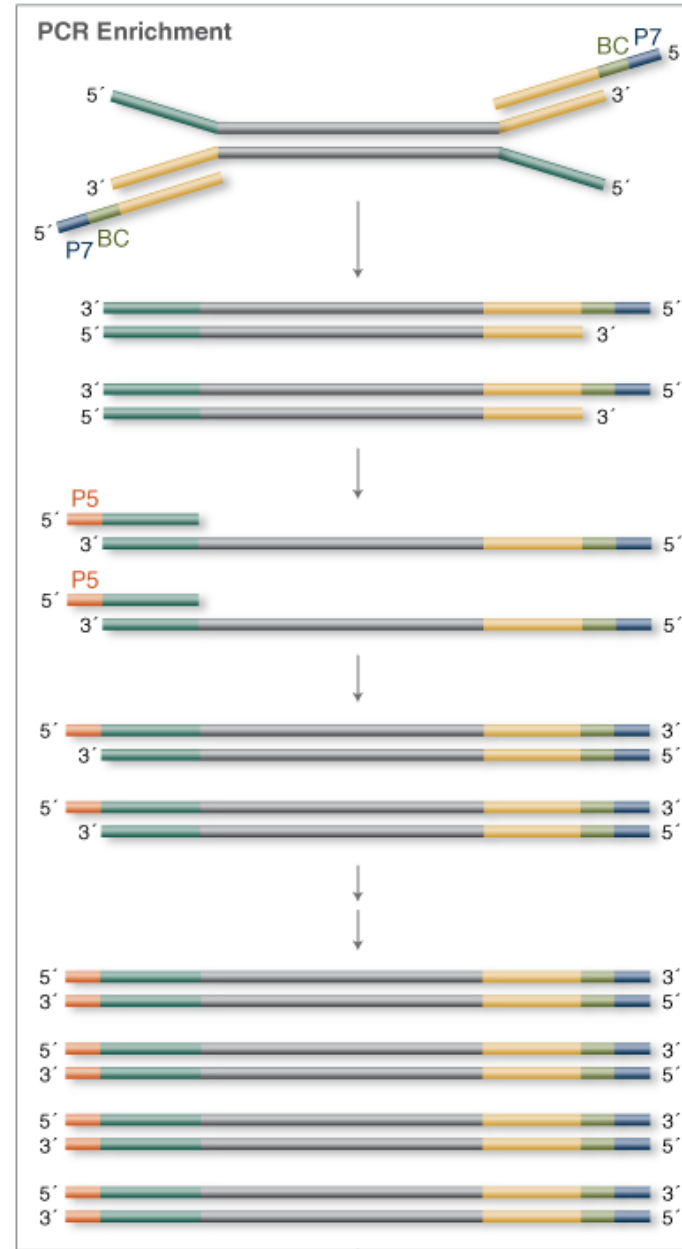
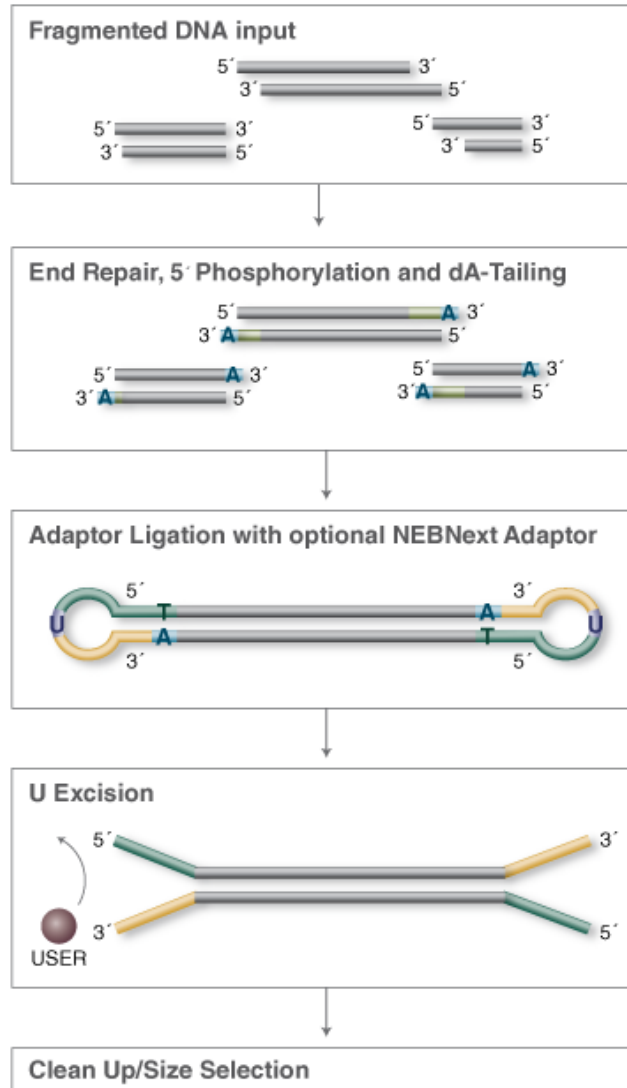


Data analysis

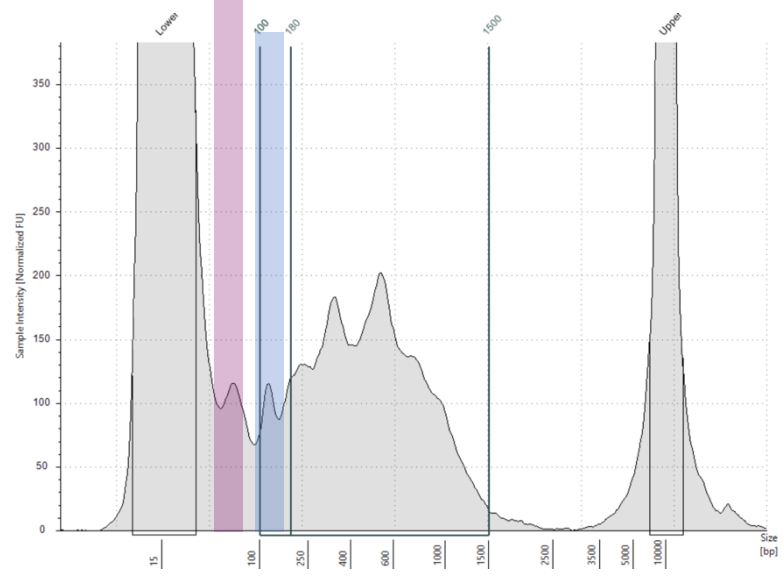
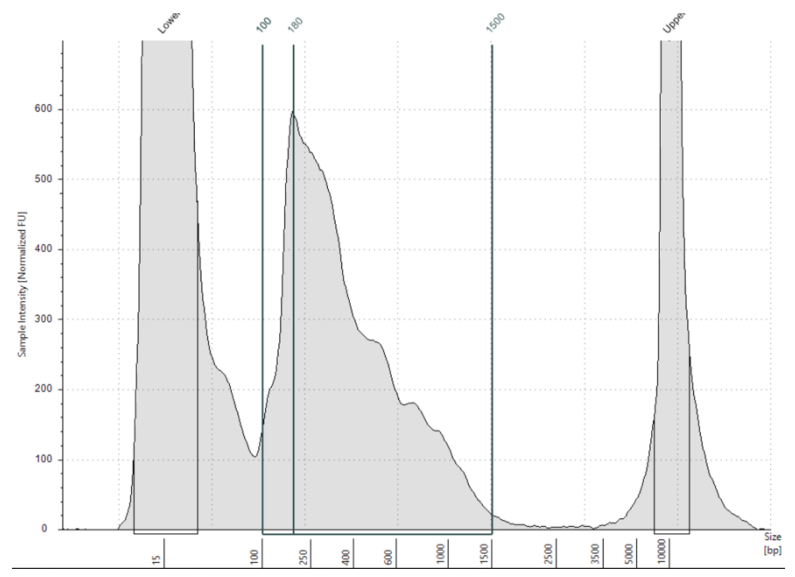
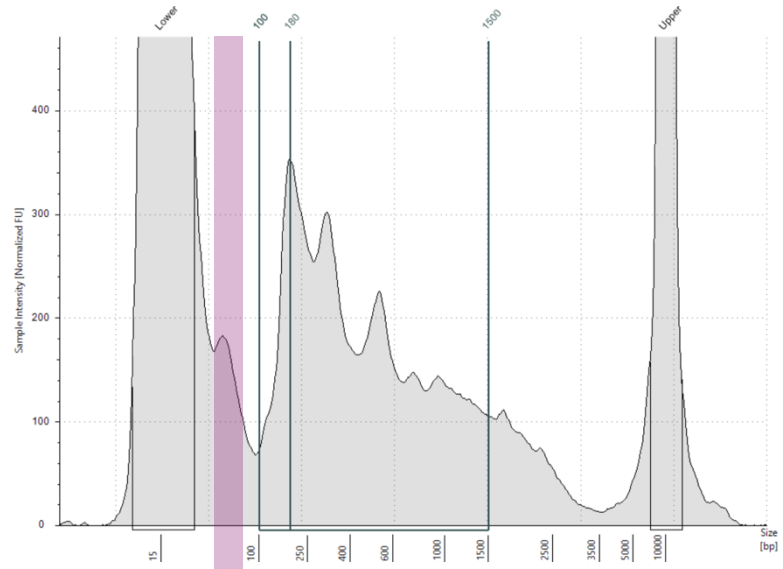
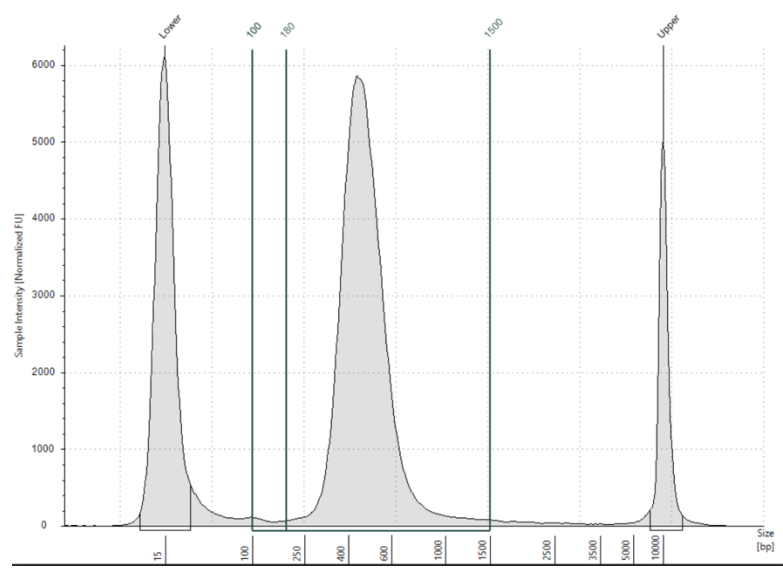
Reads are exported as an output file

Cluster 1 >	ATGCTGTAGC
Cluster 2 >	TACAGATACG
Cluster 3 >	CTAGACCAGA
...	
Cluster n >	TAGACCAGAC

Library prep – Illumina Platforms

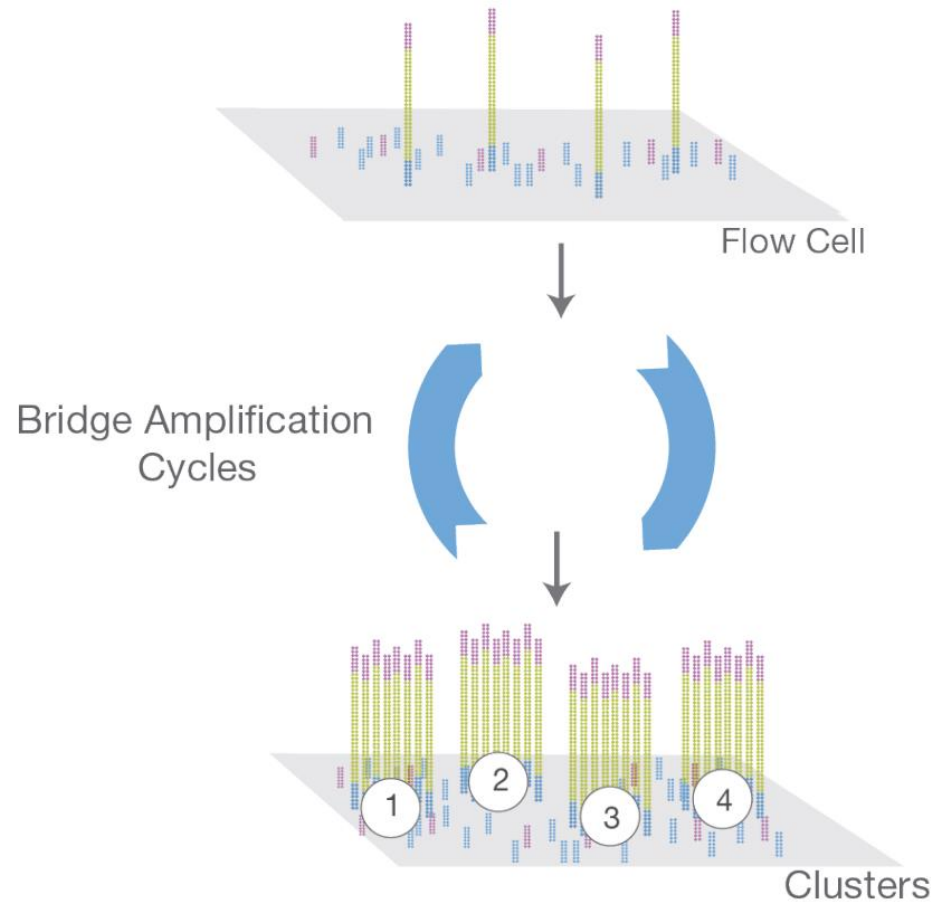


Library QC – Illumina Platforms

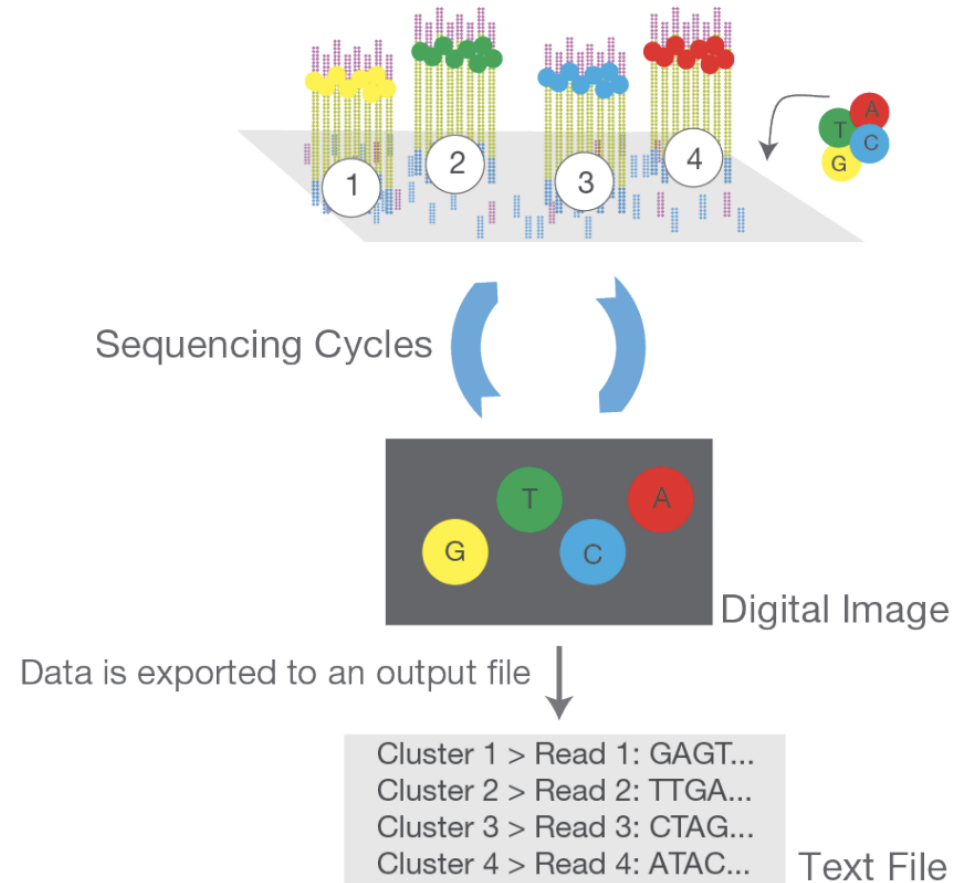


Short Read Sequencing – Illumina Platforms

Cluster Amplification



Sequencing



Run Parameters – Illumina Platforms



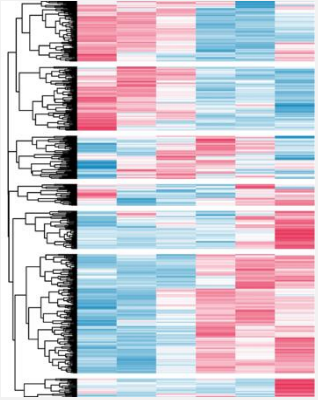
- **Read Type:** Configures the run for Single-Read (SR) or Paired-End (PE) sequencing.
- **Cycles (Read Length):** Sets the number of base pairs read per fragment. All Illumina libraries require a minimum first read length of 26 cycles.
- **Index Reads:** Defines single or dual-index configurations depending on the adapter kit.

Topics to be covered - Techniques

GENE EXPRESSION

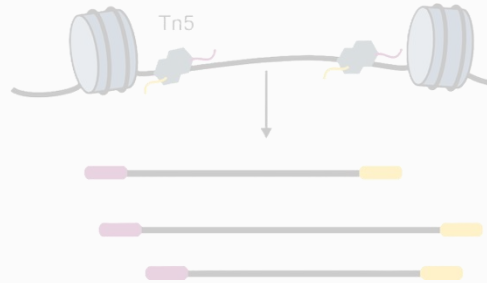
RNAseq

Spatial transcriptomics



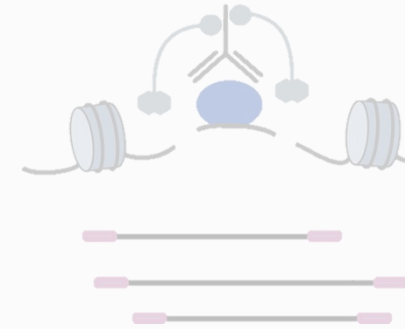
EPIGENOMIC REGULATION

ATACseq



DNA-PROTEIN BINDING

ChIPseq
CUT&RUN
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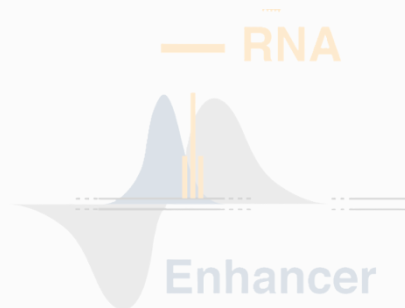
GENOME ORGANIZATION

3D genome organization - HiC



ENHANCER ACTIVITY

eRNA mapping



SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome

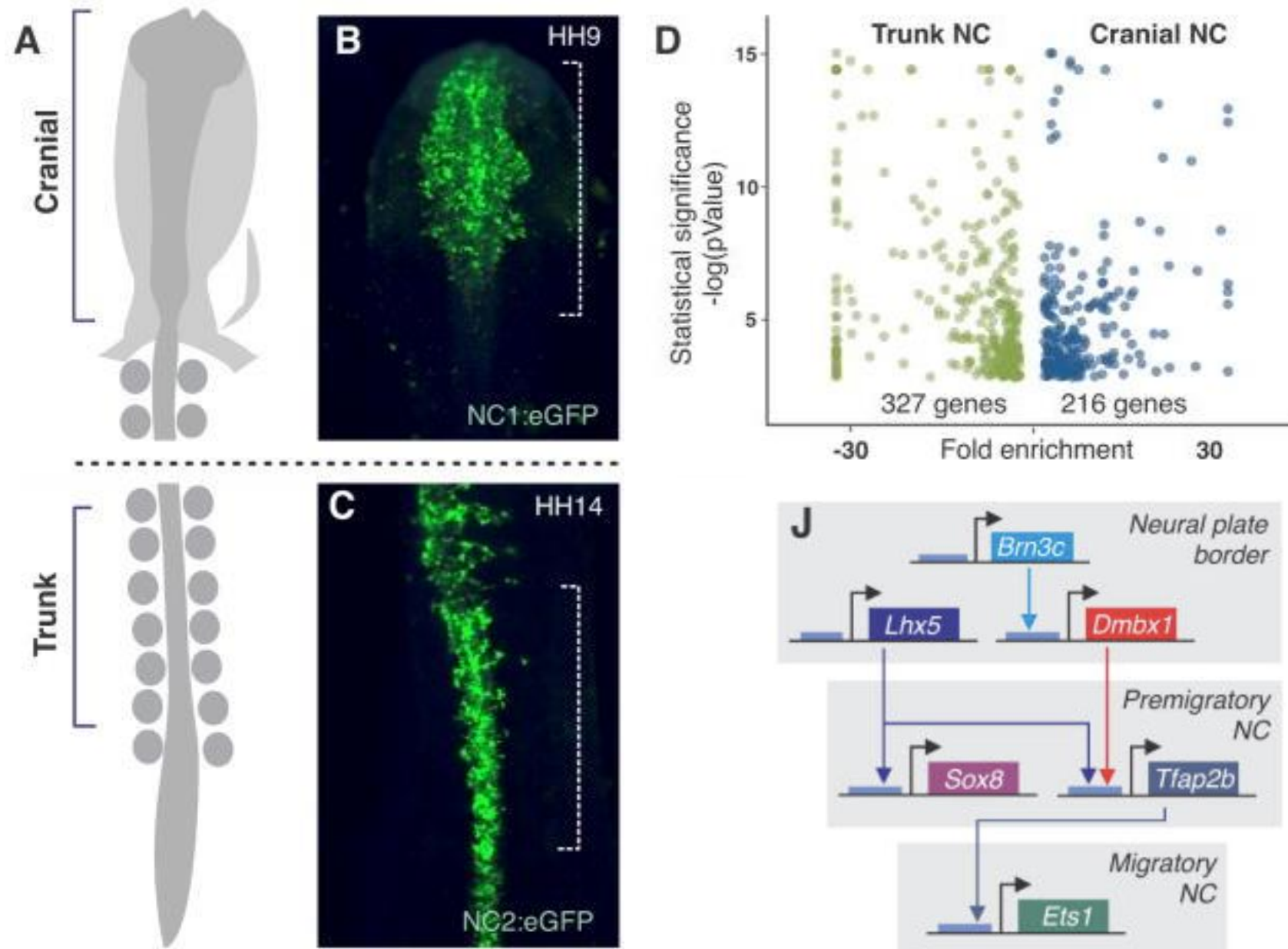


Cellular transitions are controlled by GENE REGULATORY NETWORKS



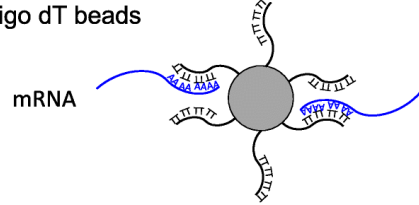
Marker genes

Identification of cranial-specific regulators by comparative transcriptomics

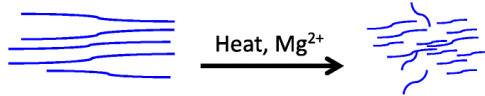


Traditional method (KAPA)

Step 1: Capture polyA tailed RNA from total RNA using magnetic oligo dT beads



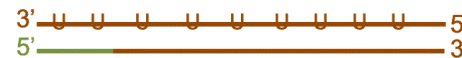
Step 2: mRNA fragmentation



Step 3: 1st strand synthesis with random primers



Step 4: 2nd strand synthesis with dUDP



Step 5: A-tailing and barcoded adapter ligation



Step 6: Amplification (dUTP strand is not amplified)



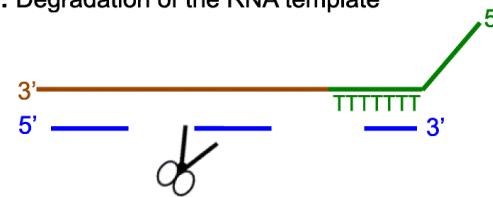
Step 7: Sequencing

3' method (LEXO)

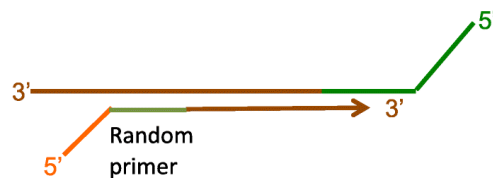
Step 1: 1st strand synthesis of polyA tailed RNA from total RNA using oligo dT primers



Step 2: Degradation of the RNA template



Step 3: 2nd strand synthesis with random primers containing 5' Illumina-compatible linker sequences



Step 4: Amplification using random primers that add barcodes and cluster generation sequences



Step 5: Sequencing

SMARTer

1. 1st strand synthesis



Template switching



2. Double-strand cDNA Synthesis

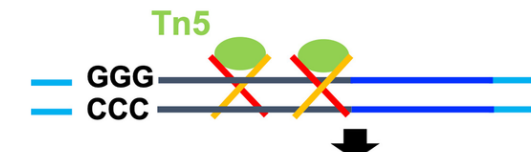


3. Amplification

Double-stranded Full-Length cDNA



4. Tagmentation



5. Sequencing



RNaseq methods

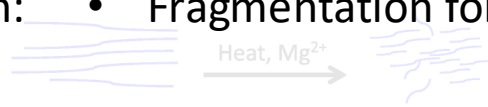
Traditional method (KAPA)

Step 1: Capture polyA tailed RNA from total RNA using magnetic oligo dT beads

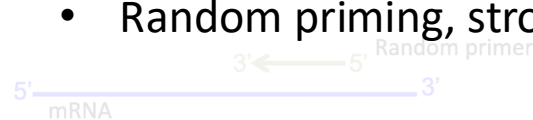
- Input:
- High-input bulk tissue (typically, 100 ng to 1 µg)



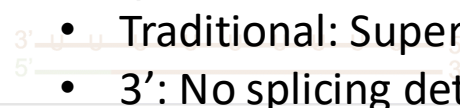
Step 2: mRNA fragmentation



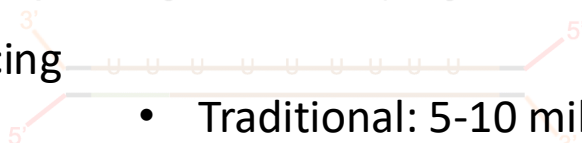
Step 3: 1st strand synthesis with random primers



Step 4: 2nd strand synthesis with dUDP



Step 5: A-tailing and barcoded adapter ligation



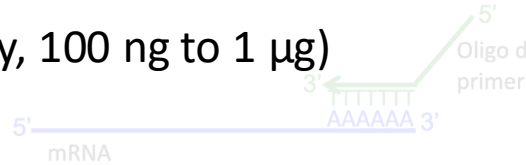
Step 6: Amplification (dUTP treatment and PCR amplified)



Step 7: Sequencing

3' method (LEXO)

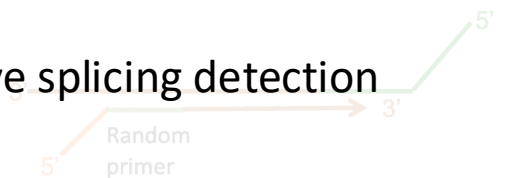
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SMARTer

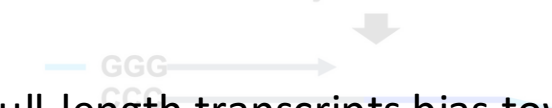
1. 1st strand synthesis



Template switching



2. Double-strand cDNA Synthesis



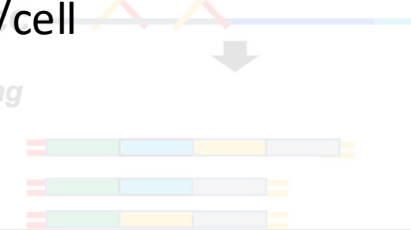
3. Amplification



4. Tagmentation



5. Sequencing



- Ultra-low input / Single-cell (10 pg to 10 ng)

- Template Switching Technology (SMART)

- Full-length transcripts bias towards 5' ends

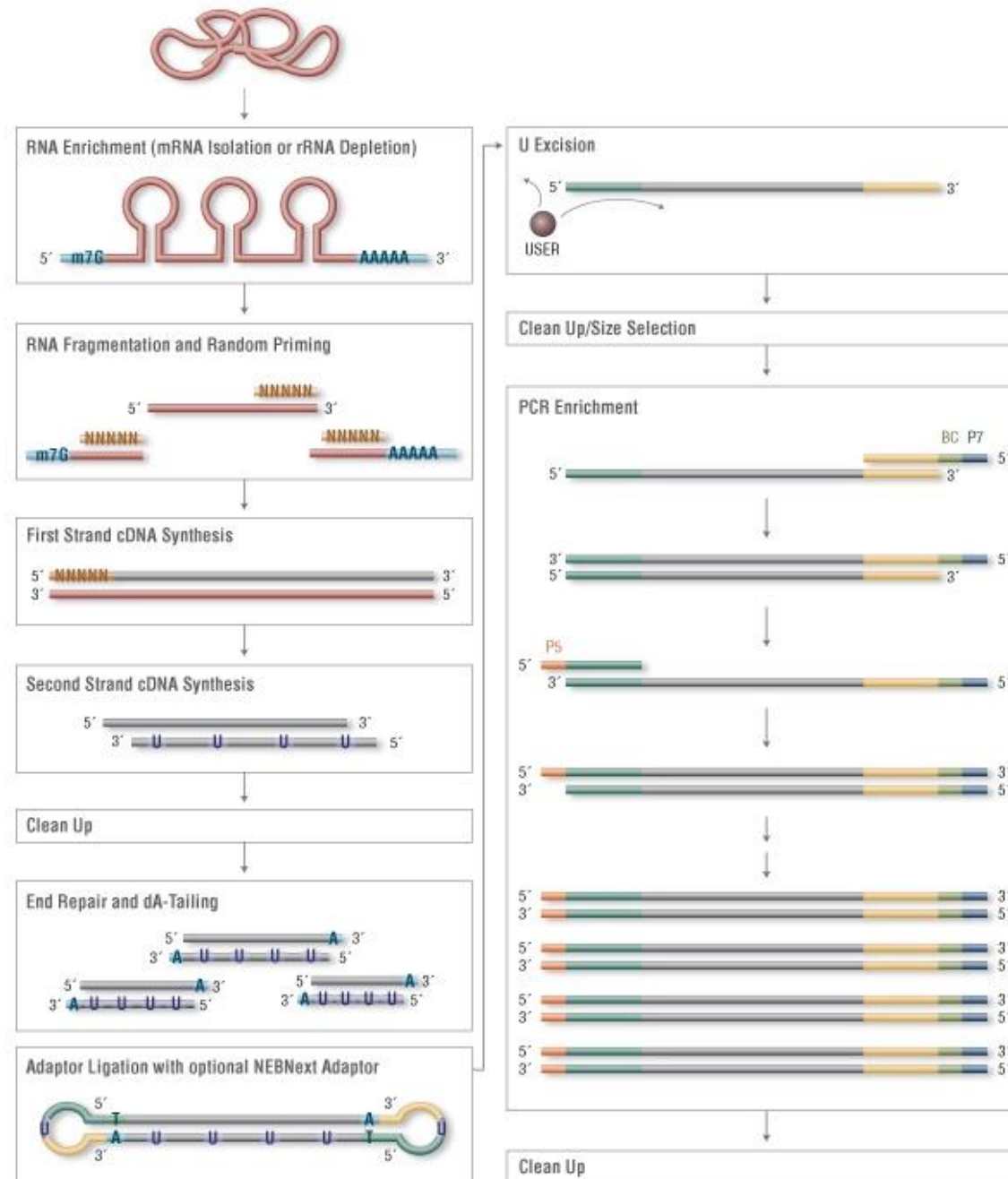
- Slightly lower alternative splicing

- 50-100k/cell

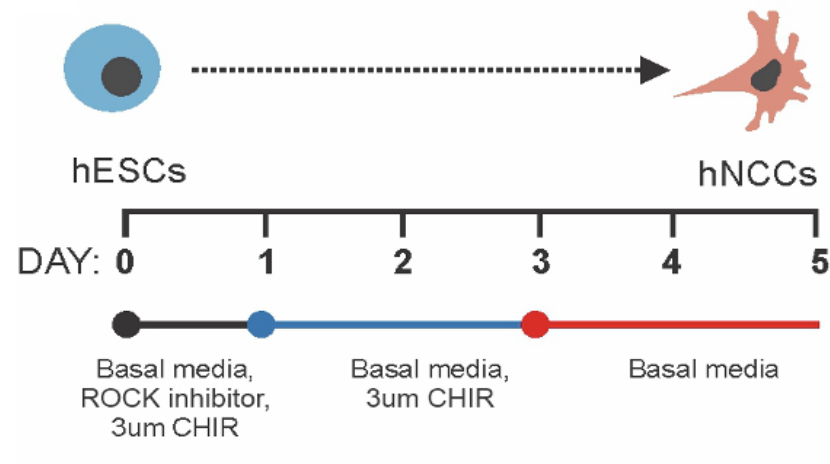
Fuqua, et al. 2019. BMC Genomics

Ura, et al.. 2022. BMC Genomics

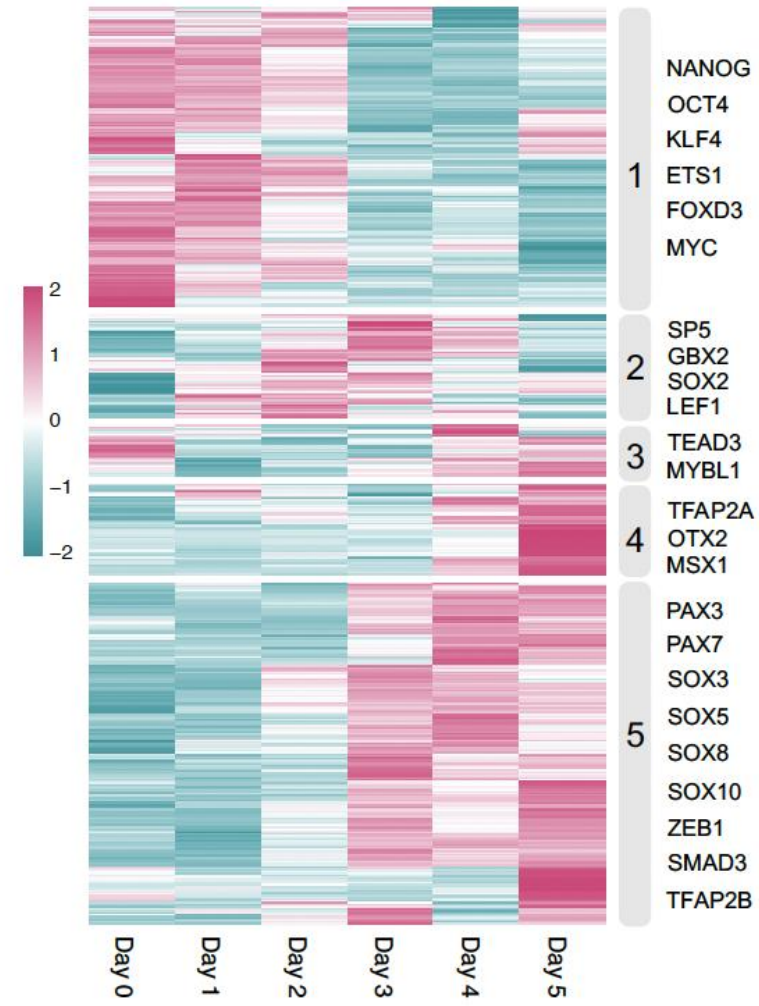
RNAseq – Library preparation



Mapping the dynamics of gene expression in a stem-cell induction protocol

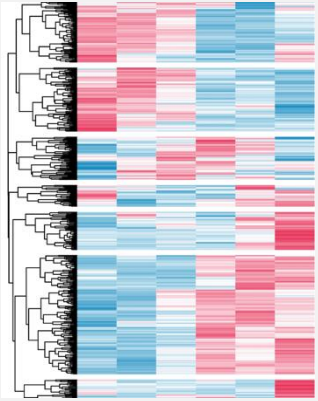


RNA extraction
↓
RNAseq lib prep
↓
sequencing
↓
differential transcriptomic analysis



Topics to be covered - Techniques

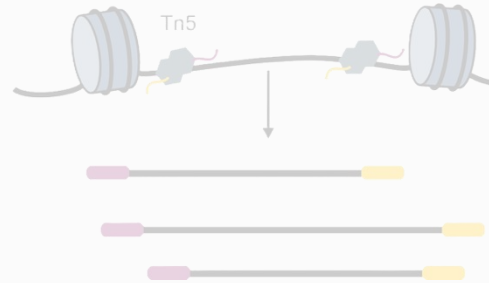
GENE EXPRESSION



RNAseq

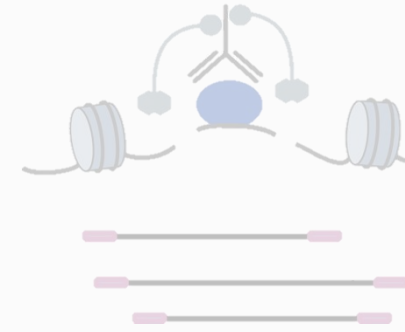
Spatial transcriptomics

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DNA-PROTEIN BINDING



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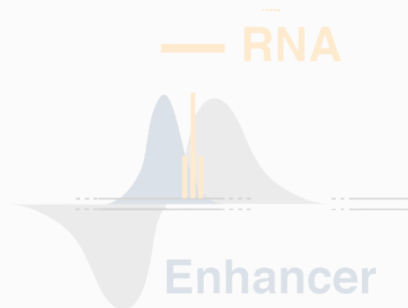
3D genome organization - HiC



ENHANCER ACTIVITY

eRNA mapping

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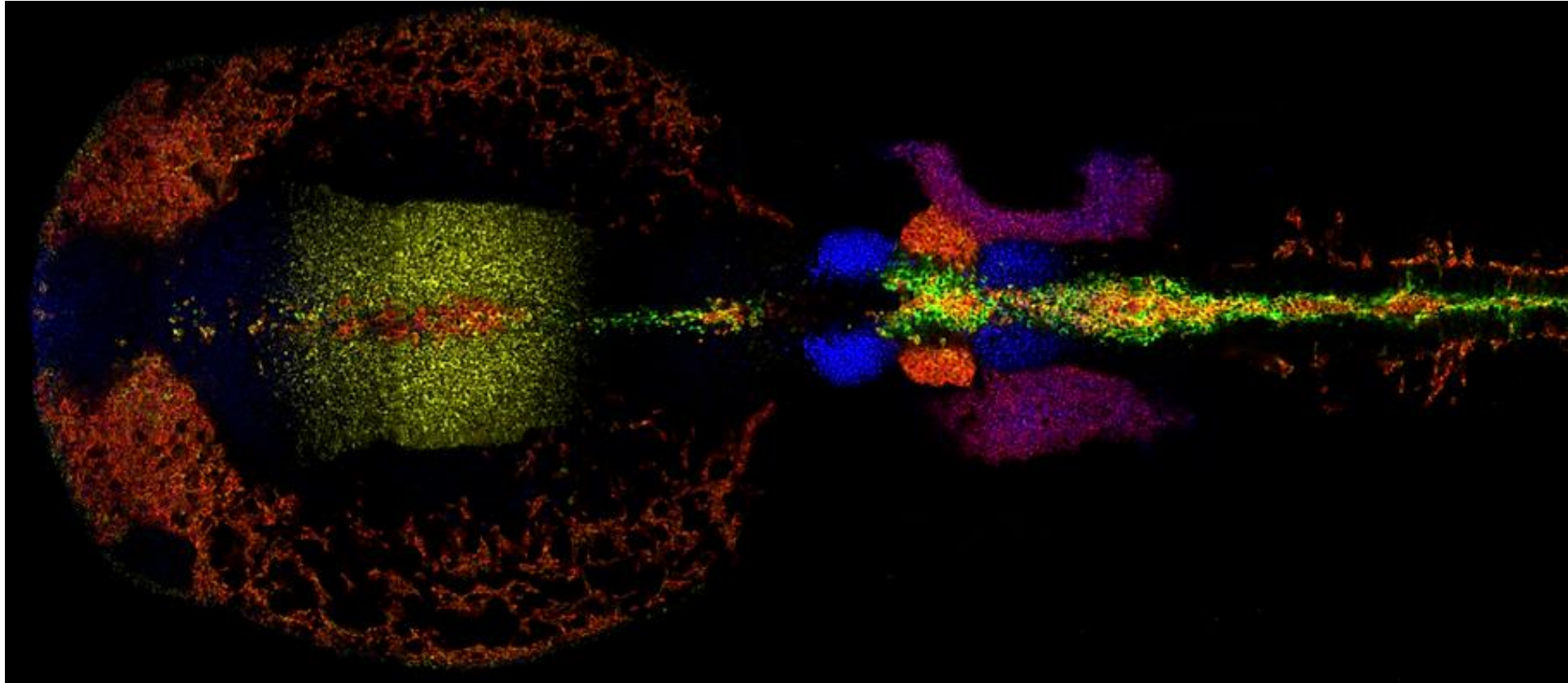


SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome



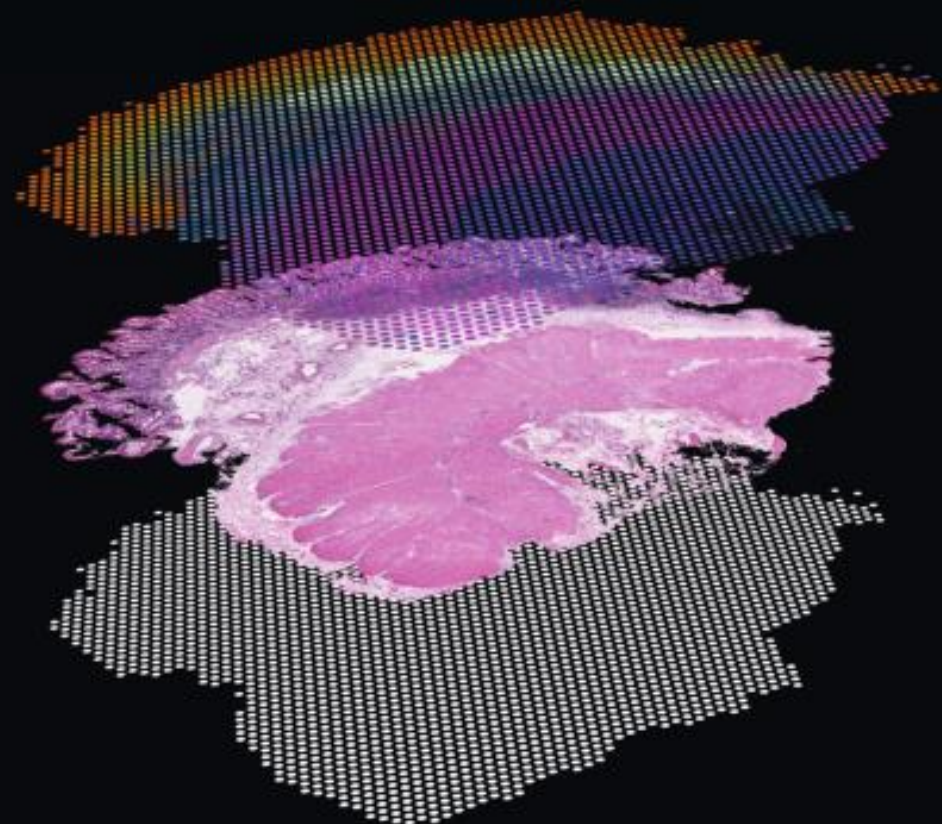
What about the spatial information?



picture from Molecular Instruments

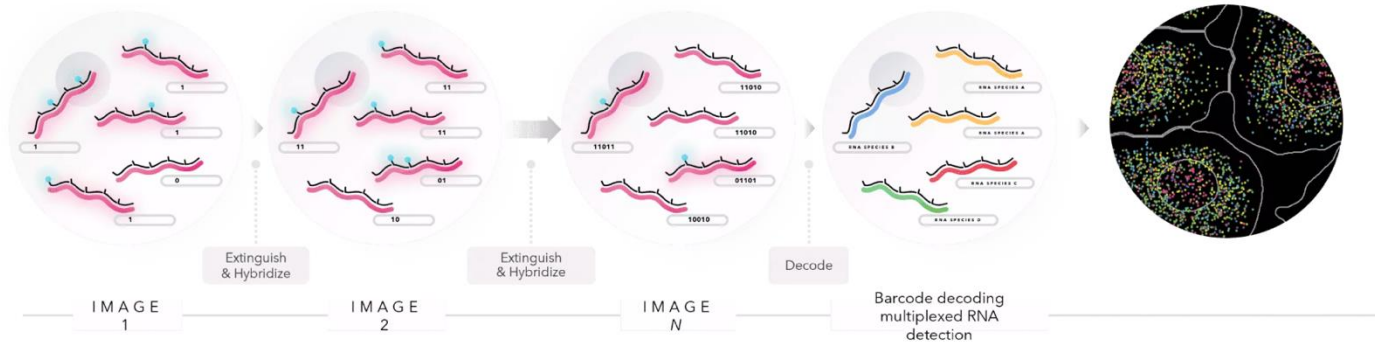
nature methods

Method of the Year 2020:
Spatially resolved transcriptomics



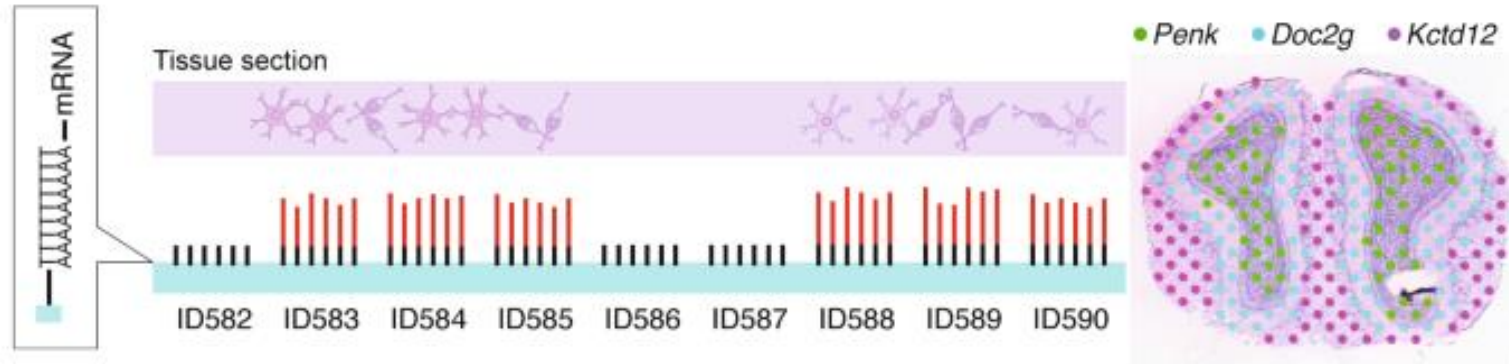
Spatial transcriptomics methods

vizgen



- imaging-based
- subcellular, single-cell resolution
- targeted gene panels

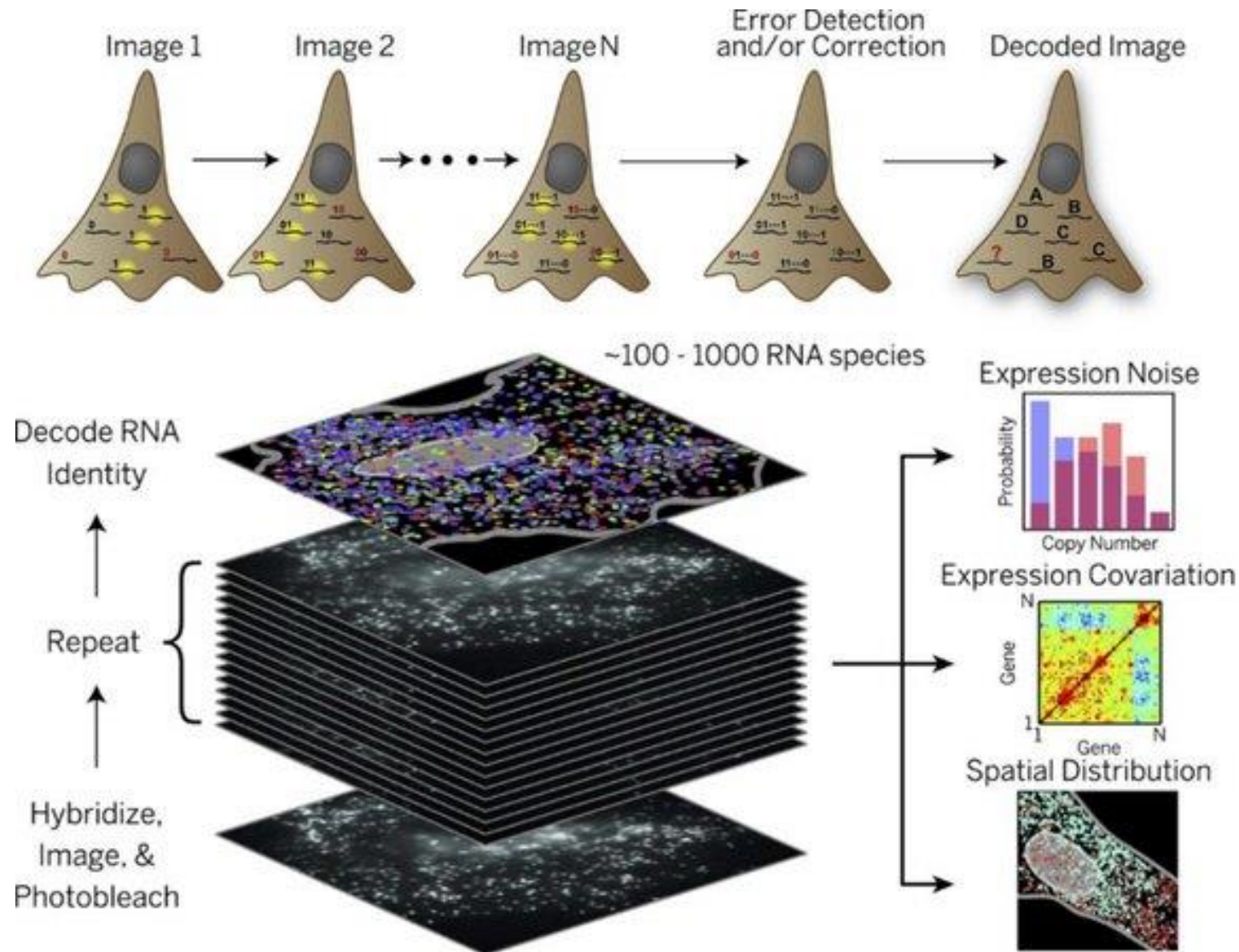
10x GENOMICS®



- sequencing-based
- unbiased, whole-transcriptome profiling
- larger tissue areas but at a **lower (multi-cell)** resolution

Spatial transcriptomics – Merfish

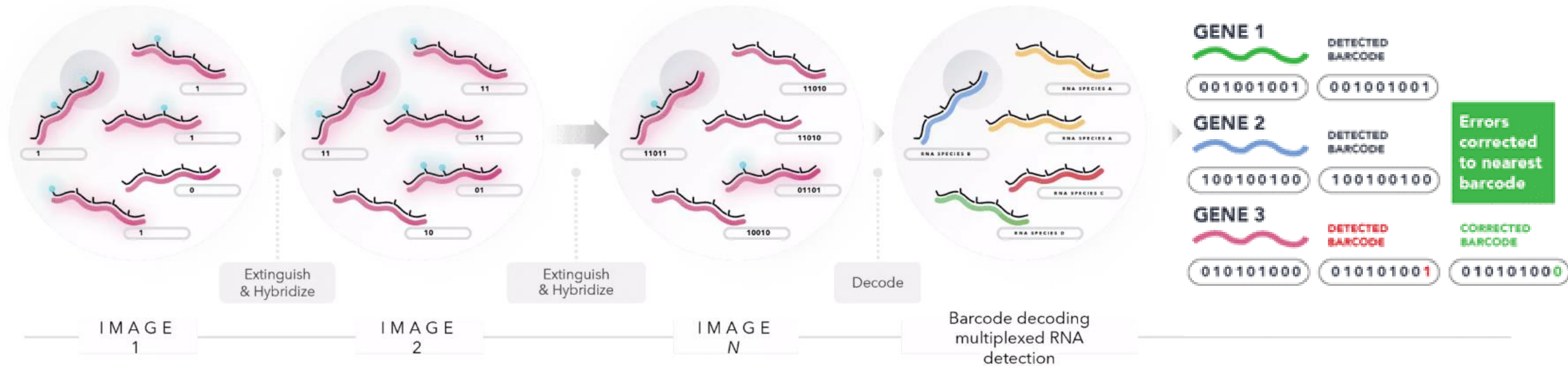
combines the principles of **F**luorescence **I**n **S**itu **H**ybridization (FISH) with **M**ultiplexed **E**rror-**R**obust barcoding strategies



- Targeted technology
- Single-cell / single-molecule
- Subcellular resolution

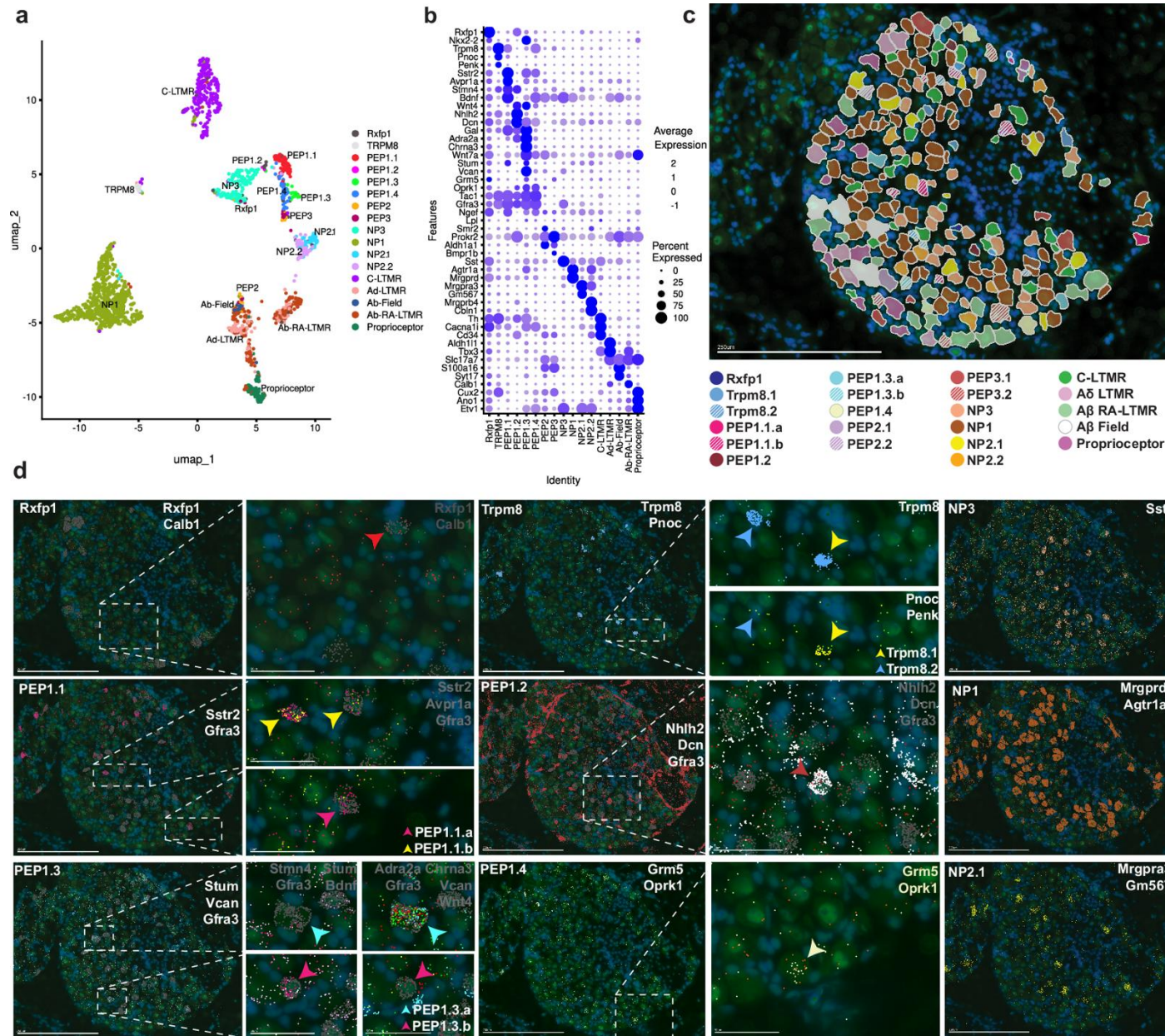
Spatial transcriptomics – Merfish

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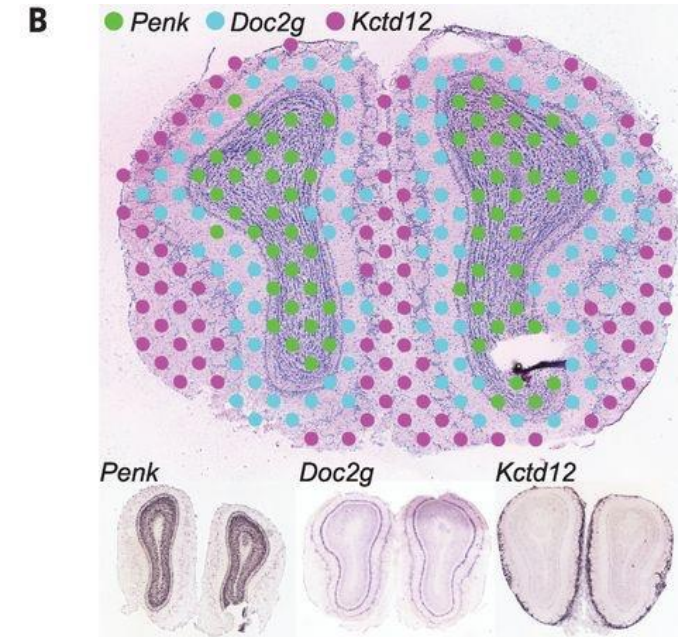
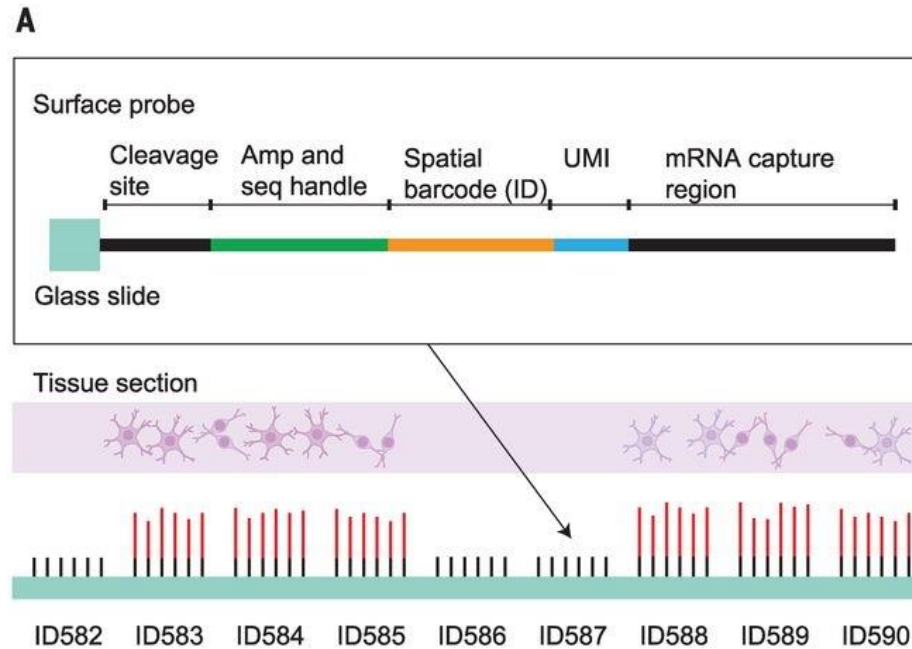
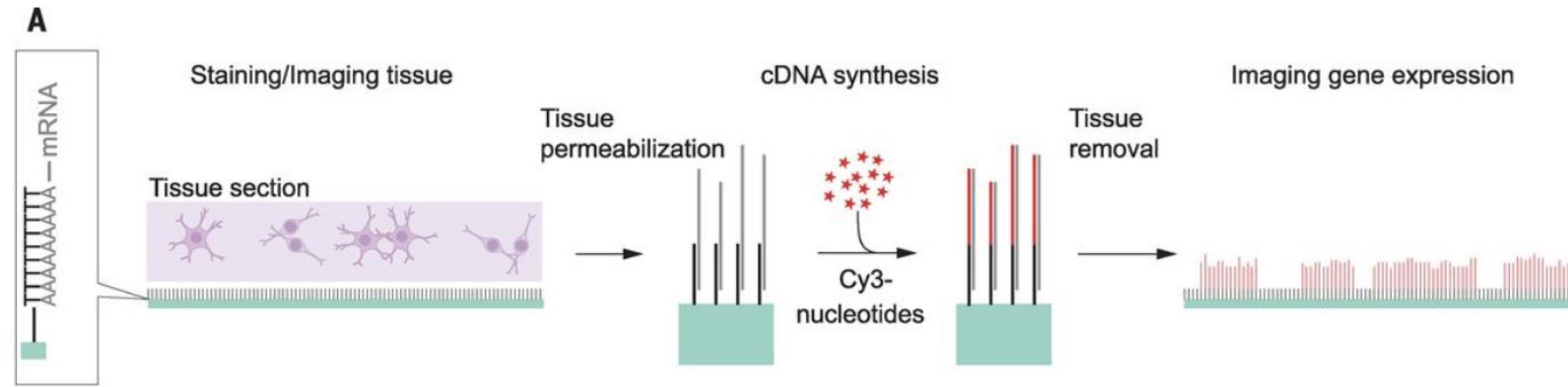


Spatial transcriptomics – Merfish

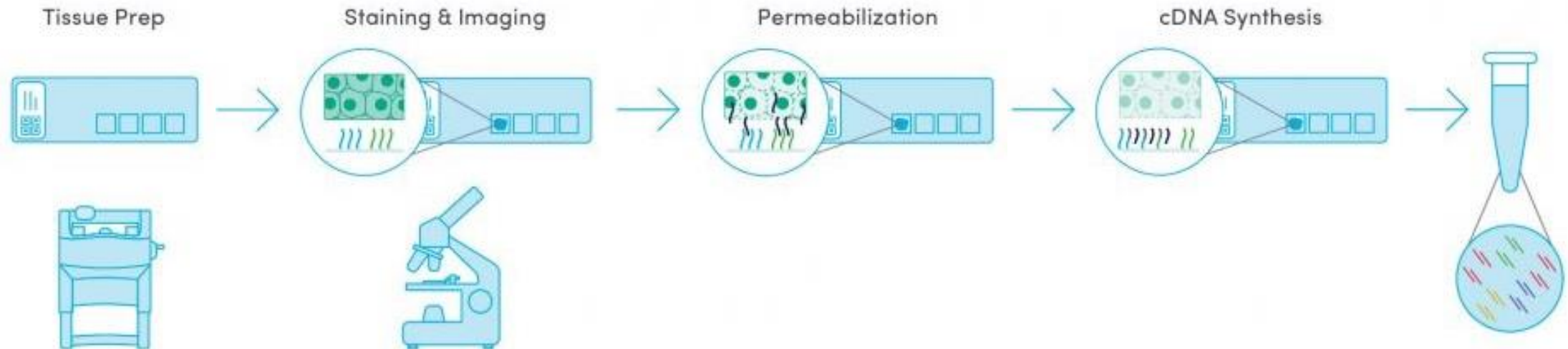
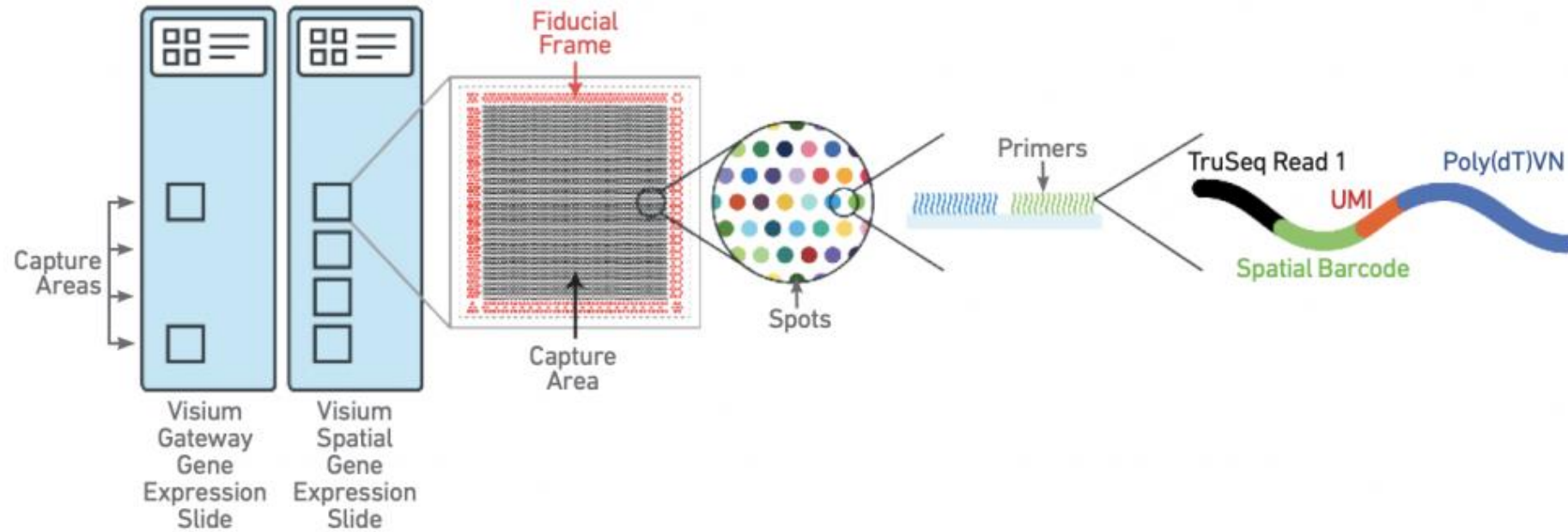
combines the principles of **Fluorescence In Situ Hybridization (FISH)** with **Multiplexed Error-Robust barcoding** strategies



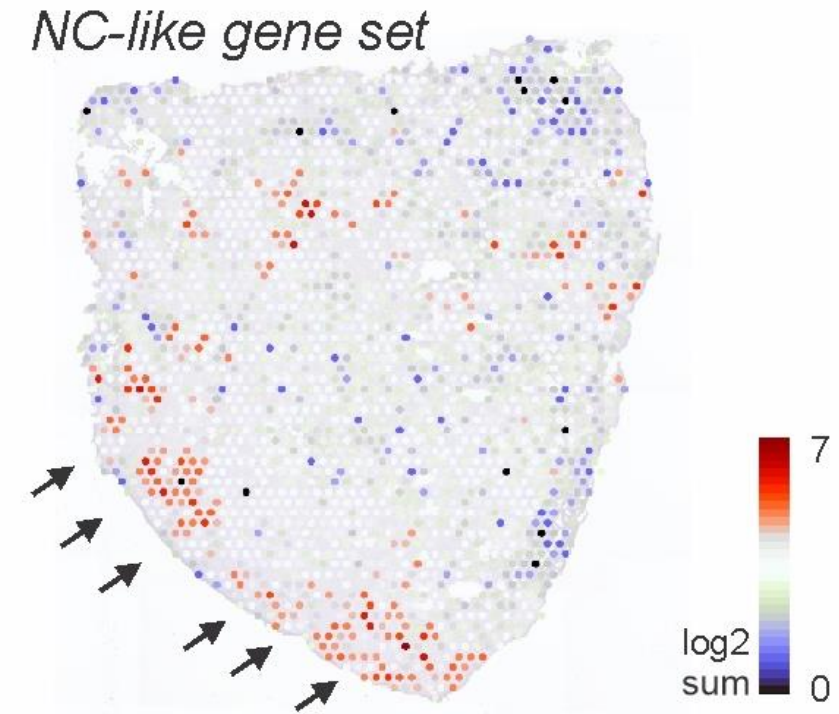
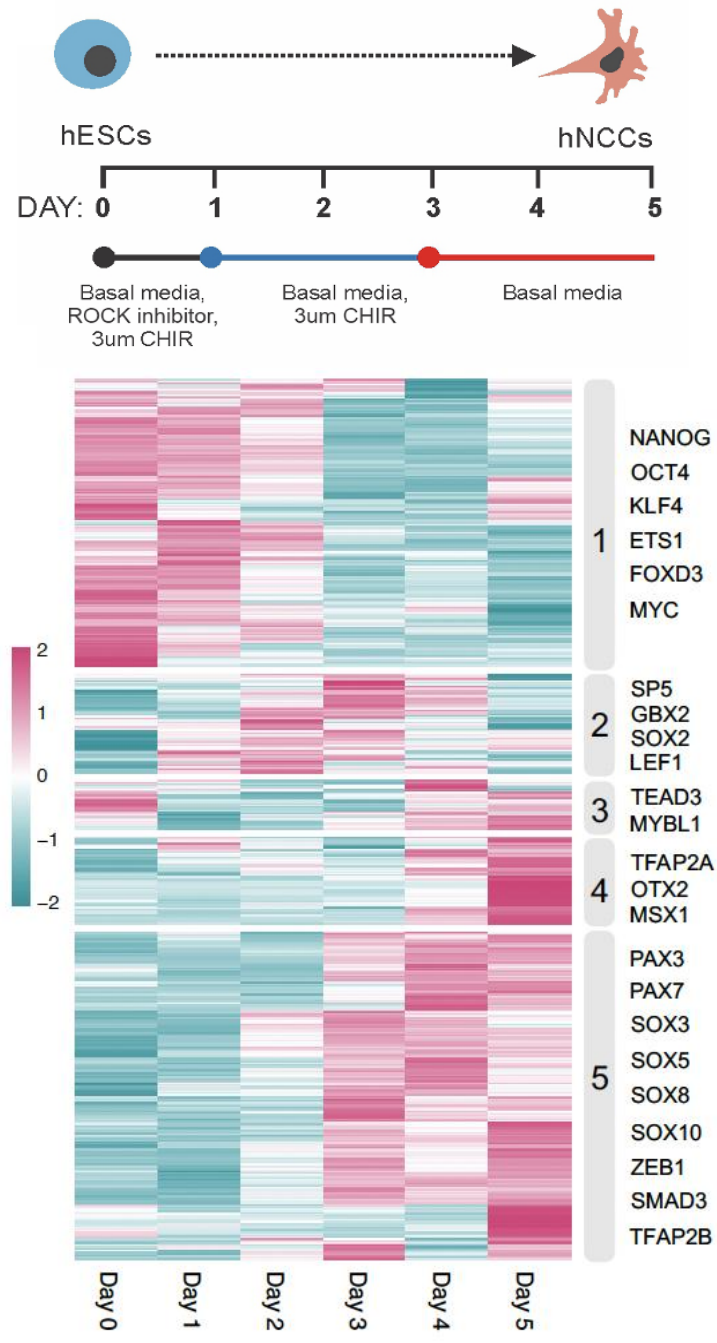
Spatial transcriptomics – 10x Visium



Spatial transcriptomics – 10x Visium



Mapping a transcriptomic signature spatially

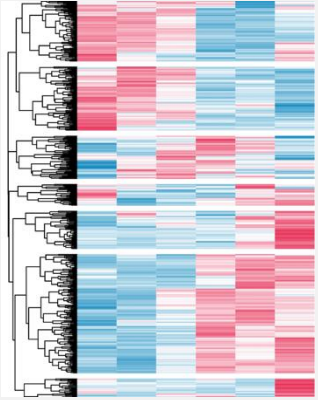


Topics covered - Techniques

GENE EXPRESSION

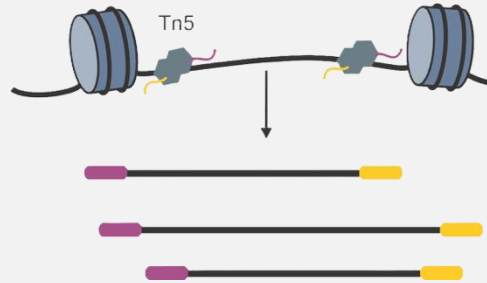
RNAseq

Spatial transcriptomics



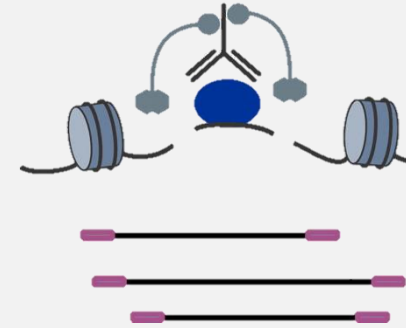
EPIGENOMIC REGULATION

ATACseq



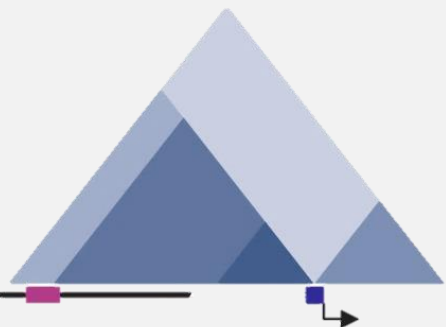
DNA-PROTEIN BINDING

ChIPseq
CUT&RUN
CUT&TAG



GENOME ORGANIZATION

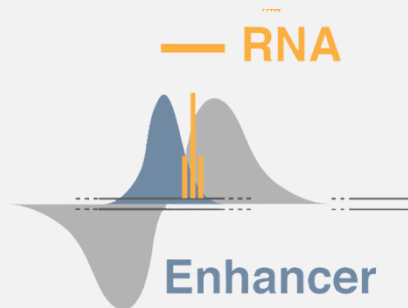
3D genome organization - HiC



ENHANCER ACTIVITY

eRNA mapping

MPRA



SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome

